



Sustainable Development Verified Impact Standard

IMPROVED COOKSTOVES DISTRIBUTION IN RWANDA

Document Prepared by Eni S.p.A.

Project Title	<i>Improved Cookstove Distribution in Rwanda</i>
Version	<i>1.0</i>
Date of Issue	<i>31 July 2024</i>
Project Location	<i>Eastern Province (Nyagatare District)</i>
Project Proponent(s)	<i>Eni S.p.A.</i> <i>Address: Via Emilia 1, San Donato Milanese (MI), Italy</i> <i>Telephone: +39 0252042086</i> <i>Website: https://www.eni.com/it-IT/home.html</i>
Assessor Contact	
Project Lifetime	<i>14th December 2023 – 13th December 2033, 10 years lifetime</i>
History of SD VISTA Status	<i>n.d.</i>
Other Certification Programs	
Expected Future Assessment Schedule	

CONTENTS

IMPROVED COOKSTOVES DISTRIBUTION IN RWANDA.....	1
CONTENTS	2
1 SUMMARY OF SDG CONTRIBUTIONS	3
2 PROJECT DESIGN.....	9
2.1 Project Objectives, Context and Long-term Viability	9
2.2 Stakeholder Engagement	24
2.3 Project Management	35
3 BENEFITS FOR PEOPLE AND PROSPERITY	40
3.1 Condition of Stakeholders at Project Start	40
3.2 Expected Impacts on Stakeholders	42
3.3 Stakeholder Monitoring Plan	45
3.4 Net Positive Stakeholder Well-being Impacts	52
4 BENEFITS FOR THE PLANET	53
4.1 Condition of Natural Capital and Ecosystem Services at Project Start	53
4.2 Expected Impacts on Natural Capital and Ecosystem Services	54
4.3 Natural Capital and Ecosystem Services Monitoring Plan	55
4.4 Net Positive Natural Capital and Ecosystem Services Impacts.....	60
APPENDIX	62

1 SUMMARY OF SDG CONTRIBUTIONS

The SDGs contributions are classified into “Direct” or “Indirect” effects of project activities.

For the SDGs directly impacted by the project (1,3,5,7,8,13), the SD VISta labeled VCUs for VCS project ID 5196 is required: specific indicators will be measured and monitored during project lifetime for each SDG.

For the SDGs indirectly impacted by project activities (2,4,10,15,17), the SD VISta claim for VCS project ID 5196 is required: they will be supported by qualitative and literature research analysis.

In the following table, a list of specific indicators for all SDGs, both directly and indirectly impacted by project activities, is showed. For each SDG “directly” impacted by project activities, at least one indicator is measured and monitored and may be linked to the different SDG targets identified. In the column “section reference” it is indicated the section in the document where the project indicators/parameters for the monitoring of the contribution to the SDGs are described: this column relates only to the SDGs indicators considered to be “directly” impacted *by the project’ activities*.

Table 1: Summary of Project SDG Contributions

Row number	Estimated Project Contribution by the End of Project Lifetime	SDG Target	SDG Indicator	Net Impact on SDG Indicator	Section Reference	Claim, Asset or Label
1)	SDG1 The project contributes to the eradication of extreme poverty, currently measured as people living on less than \$1.25 per day: indeed, the distribution of improved cookstoves allows the communities to reduce the biomass	1.1	1.1.1. Proportion of the population living below the international poverty line by sex, age, employment status and geographic location (urban/rural)	<u>Implemented activities to</u> Decrease INDIRECT EFFECT of the project	<u>3.2 #2 #3 #4</u>	SD VISta labeled VCUs

	collection/purchase, with a direct impact on the household's budget.					
		1.4	Indicator 1.4.1: Proportion of population living in households with access to basic services	Increase DIRECT EFFECT of project activities	3.2 #1	
2)	SDG2 The access to an Improved cookstove technology allows household to increase food security and nutrition intake by saving budget, resources and time.	2.1	Indicator 2.1.1: prevalence of undernourishment.	<u>Implemented activities to</u> Decrease INDIRECT EFFECT of project activities	<u>3.2 #4</u>	SD VISta claim
		2.2	Indicator 2.2.2: prevalence of malnutrition among children under 5 years of age, by type (wasting and overweight).	<u>Implemented activities to</u> Decrease INDIRECT EFFECT of project activities	<u>3.2 #4</u>	
3)	SDG3 The use of less polluting fuels and technologies improve health condition among households, especially respect to the respiratory illness due to indoor air pollution and to the risk of	3.2.	Indicator 3.2.1. Under 5 morality rate	<u>Implemented activities to</u> Decrease	3.2 #1	SD VISta labeled VCUs

	young children to fall into open cooking fires			INDIRECT EFFECT of project activities		
		3.9	Indicator 3.9.1: Mortality rate attributed to household and ambient air pollution	Decrease DIRECT EFFECT of project activities	3.2 #2	
4)	SDG4 Children, particularly girls, are often kept out of school so that they can contribute to household tasks, like cooking and collecting fuel.	4.1	Indicator 4.1.2: Completion rate (primary education, lower secondary education, upper secondary education)	<u>Implemented activities to</u> Increase INDIRECT EFFECT of project activities	<u>3.2 #3</u>	SD VISTA claim
		4.7	Indicator 4.7.1: Extent to which (i) global citizenship education and (ii) education for sustainable development are mainstreamed in (a) national education policies; (b) curricula; (c) teacher education; and (d) student assessment	<u>Implemented activities to</u> Increase INDIRECT EFFECT of project activities	3.2 #7	
5)	SDG5 Women have the responsibility for domestic tasks in the households, including fuel provisioning and cooking. The use of improved cookstoves would	5.4	Indicator 5.4.1: Proportion of time spent on unpaid domestic and care work, by sex, age and location	Decrease	3.2 #3	SD VISTA labeled VCUs

	reduce the practices that are very time consuming. The possibility to save time can engage women in other income generating activities or pursue educational opportunities, that are some of the causes of gender inequality.			DIRECT EFFECT of project activities		
6)	SDG7: Increasing access to more efficient and cleaner cooking technology allows to address energy poverty and ensure sustainable energy security	7.1	Indicator 7.1.2: Proportion of population with primary reliance on clean fuels and technology	Increase DIRECT EFFECT of project activities	3.2 #1	SD VISta labeled VCUs
7)	SDG8: Clean cooking sector offers many job opportunities, both with the direct employment in the cookstove production and distribution and the indirect employment given by the budget saving of each household.	8.5	Indicator 8.5.2: Unemployment rate, by sex, age and persons with disabilities	Decrease DIRECT EFFECT of project activities	3.2 #5	SD VISta labeled VCUs
		8.2	Indicator 8.2.1: Annual Growth rate of real GDP per employed person	Increase DIRECT EFFECT of project activities	3.2 #5	
8)	SDG10: Reduce the disparity in energy access and technology is key in order to reduce income inequalities, gender inequalities and rural/urban income disparities within a Country and to	10.2	Indicator 10.2.1: Proportion of people living below 50 per cent of median income, by sex, age and persons with disabilities	<u>Implemented activities to decrease</u>	3.2 #4	SD VISta claim

	increase opportunities in the most remote areas			INDIRECT EFFECT of project activities		
9)	SDG13: The less polluting cooking technology and the decreasing rate in deforestation are both actions to deal with climate change impacts.	13.2	Indicator 13.2.2: Total greenhouse gas emissions per year	Decrease (as compared to baseline scenario) DIRECT EFFECT of project activities	4.2 #1	SD VISta labeled VCUs
		13.3	Indicator 13.3.1 Extent to which (i) global citizenship education and (ii) education for sustainable development are mainstreamed in (a) national education policies; (b) curricula; (c) teacher education; and (d) student assessment	<u>Implemented activities to increase</u> INDIRECT EFFECT of project activities	<u>3.2 #7</u>	
10)	SDG15: The improved cookstove allows to consume less biomass and therefore the project contributes in reducing the deforestation rate and the pressure on forest reserve	15.1	Indicator 15.1.1: Forest area as a proportion of total land area	<u>Implemented activities to increase</u> (as a trend) compared to baseline scenario	<u>4.2 #2</u>	SD VISta claim

				INDIRECT EFFECT of project activities		
		15.2	Indicator 15.2.1 Progress towards sustainable forest management	<u>Implemented</u> <u>activities to</u> <u>increase</u> INDIRECT EFFECT of project activities	<u>4.2 #2</u>	
11)	SDG17 Goal 17. Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development: the project activities encourage and promote effective public, public-private and civil society partnerships, building on the experience and resourcing strategies of partnerships	17.17	17.17.1 Amount in United States dollars committed to public-private partnerships for infrastructure	<u>Implemented</u> <u>activities to</u> <u>increase</u> INDIRECT EFFECT of project activities	<u>3.2 #7</u>	SD VISta claim

2 PROJECT DESIGN

2.1 Project Objectives, Context and Long-term Viability

2.1.1 Summary of Project Sustainable Development Objective(s)

The distribution of improved cookstoves is one of the activities that may accelerate the access to “clean cooking”, an urgent development issue with significant benefits for public health, gender equality, local environment and global climate.

The specific effects of project activities that will lead to expected SD contributions are:

- Reduction in the consumption and use of biomass fuels for cooking activities;
- Reduction in the time spent for cooking and fuel-collection activities;
- Engagement and training process of local communities on health, safety and environmental issues.

All these activities are contributing to the achievement of sustainable development goals with respect to the baseline scenario.

In order to increase the accountability rate of the project with respect to the achievement of the SDGs, the contributions have been split into “Direct” and “Indirect” effects of the project activities. The “Direct” contributions are monitored through quantitative analysis and surveys, as described in sections 3.2 and 4.2; the “Indirect” contributions are supported by qualitative and literature research in the area and are not directly measured in the field.

DIRECT Contribution to the achievement of sustainable development goals by the project activities:

- **SDG1: End Poverty in all its form everywhere**

In this project, each household receives the cookstove free of charge. Considering the current cooking fuel utilization, the distribution of improved cookstoves increases household income by reducing the amount of wood and charcoal purchased every month. Moreover, by reducing the amount of time spent collecting firewood, household time is freed up for activities to generate additional income. Local communities involved in the project are also going to be engaged by receiving dedicated training on cookstove utilization so that they can acquire how to optimize the time and resources for cooking activities.

- **SDG3: Ensure healthy lives and promote well-being for all at all ages**

The use of polluting fuels and technologies in traditional ways of cooking is causing respiratory illness, heart problems, and even death in Sub-Saharan African Countries: household air pollution (HAP) contributes to at least 581,000 premature African deaths per year and the loss of more than 26 million disability-adjusted life years (DALYs), out of a global total of 4.3 million deaths (2012) and 110 million HAP DALYs (2010) [1 (*Kammila, 2014*)]. According to World Bank (2017) [2], the Rwandan population still relies on wood or charcoal for more than 80% of its energy needs for cooking. According to the World Bank, the life expectancy in Rwanda is 66 years in 2021¹. Indoor air pollution due to the use of inefficient cooking appliances caused more than 24,500 deaths in 2019. [3]

- **SDG5: Achieve gender equality and empower all women and girls.**

The energy sector in sub-Saharan Countries has been widely studied from a gender perspective. The report “The state of gender equality in Rwanda: from transition to transformation” (2019) [4] highlights how women are mostly engaged in farming and domestic activities.

The women have the responsibility for domestic tasks, including fuel provisioning and cooking, experiencing the highest opportunity costs (less time for education, leisure, and remunerative activities). The use of improved cookstoves would reduce the current practices, which are very time-consuming (5 hours per day) are spent on average for cooking and collecting firewood [5]. The possibility to save time can engage women in other income-generating activities, pursue educational opportunities, or simply rest. Moreover, some studies find a relationship between the gender base violence during the firewood collection activities and the women’s exposure to assault during the collection trips [6].

- **SDG7: Ensure access to affordable, reliable, sustainable and modern energy for all**

Clean cooking activities are crucial to reach the SDG7. Recently, the International Energy Agency has dedicated a special report on this critical issue (“A vision for clean cooking access for all”, 2023) dedicated to *“the current state of clean cooking, the cost of inaction and the benefits of boosting access in line with SDG7”* [7].

According to the World Bank database, in 2020 the population’ share with access to modern energy cooking services in Rwanda was only 5% [8]. This value goes down to 0.7% considering the rural population [9]. In the report “State of Access to Modern Energy Cooking Services” (World Bank, 2020), the main obstacles are a combination of underdeveloped infrastructures, high up-front capital costs and low households’ awareness. In the past decades, the interventions to increase energy access have not been sufficient to keep the path with population growth: therefore, the number of households accessing electricity has grown, but also the absolute number of people cooking with traditional stoves that burn biomass inefficiently has increased. The distribution of ICSs with a thermal efficiency more than double compared to the traditional cooking mechanisms allows the local communities to reduce the biomass collection/purchase, with a direct impact on the households’ budget.

- **SDG8: Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all.**

Within this project, the distributed improved cookstoves are coming from local manufacturers. The local production allows not only to reduce emissions by optimizing the logistic transport of each device but also to share technological knowledge, improve the local economy and generate job opportunities in clean cooking sector. Moreover, field Studies from the Global Alliance for Clean Cookstoves show that the additional time savings from collecting fuel and cooking lead to greater time for income-generating opportunities, and that this is true, especially for women. Published research from the field [10] shows a virtuous link between women and clean cooking jobs: women are often engaged as cookstove entrepreneurs since it is recognized that women’s networks can provide access to consumers in hard-to-reach markets and women distributors can better understand the needs of women and more easily approach their clients.

- **SDG13: Take urgent action to combat climate change and its impact.**

The less polluting cooking technology and the decreasing rate of deforestation are both actions to deal with climate change impacts. The improved cookstoves distributed are single-pot charcoal stoves with

thermal efficiency higher than 25%: the use of these devices will allow a GHG emission reduction compared with the current cooking methodologies due to the higher thermal efficiency. The estimated annual emission reduction for the grouped project activity is on average 25,000 tons of CO₂e per year, as estimated in the VCS project for the corresponding SD VISta period.

INDIRECT contribution to SDG achievement through the project activities

- **SDG2: End hunger, achieve food security and improved nutrition and promote sustainable agriculture.**

According to the World Food Programme [11] - the international organization within the United Nations that provides food assistance worldwide- access to clean, modern, sustainable and affordable energy plays a critical role in World Food Programme's mandate to achieve zero hunger. First, the use of more efficient cooking devices allows to increase food security due to nutrient retention with decreased cooking time. Moreover, improved cookstoves and food security are also indirectly linked by the time benefits and income gains from using these more efficient devices. Finally, considered that food stability means that households have access to sufficient and adequate food "on a continuous basis", an improvement in the efficiency of the devices contributes to guarantee that the food that needs to be cooked before consumption is prepared all the time.

- **SDG4: Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all.**

In rural areas, children are usually in charge of domestic household tasks like collecting fuel for cooking activities. These activities are time-demanding and compromise constant attendance at school. Many studies indicate that fuel scarcity places major demands on women and children's time: the EnDev "Report on impact - 2016" [12] states that collecting firewood can lead to lack of time for other purposes such as school attendance and studying; families from Kibaha District in Tanzania mentioned that their daughters were missing schools up to three days per month due to firewood collection [13]; furthermore, UNDP 2013 [14] states that an increase in the fuel wood collection time may force girls to drop out of schools to assist their mothers in household's tasks, thus hindering their education.

Furthermore, cooking activities are not free of risk, especially for children and young people. From this point of view, the implementation of this project improves children's condition, freeing up time given the efficiency of the device that requires less fuel for the same amount of thermal energy produced. The improvement is also on the safety side, reducing the burn and fire risks: however, as it explains during the implementation of the project, cooking activities shall always be performed by adults with any kind of cooking technology since they are not made for children.

- **SDG10: Reduce inequalities within and among Countries.**

This project takes place in the Eastern area of Rwanda. According to World Bank, 52% of population in Rwanda live in poverty. Despite the economic growth of 7.6 percent in the first half of 2023, per capita GDP growth declined from 8.3 (2021) to 5.7 percent (2022). As highlighted by Word Bank, despite challenging external and domestic factor, the GDP growth is expected to regain momentum in 2024-26, with a projected average growth of 7.2 percent [15]. Access to a form of clean cooking can reduce the inequality within the Country, in terms of household budget, health, and life expectation. Inequality is greater at the urban level. According to World Bank, inequality within the country is decreasing, and this is confirmed by the trend in the Gini index: the index went from 0.52 in 2005 to 0.43 in 2016, with an overall decrease of 8,3 percent.

- **SDG15: Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification and halt and reverse land degradation and halt biodiversity loss**

According to the World Bank statistical data, non-renewable biomass has been used in the project region for years. According to data, the forest area has been declining since 1990. Between 1990 and 2010 Rwanda has lost a total of 210,000 hectares, and between 2010 and 2023 it lost 1,210 ha of natural forest, an average of 6,400 hectares on yearly basis. The use of wood fuel for cooking purpose could be address as one of the main drivers of deforestation in the Country: according to World Bank Data, the local population that in 2017 was still relying on woody biomass (wood and charcoal) was 79,9%, with different proportion for urban and rural areas. Due to the improved cookstoves efficiency and the widespread community use, there will be a reduction in firewood cutting from live trees resulting in better management of protected areas.

- **SDG17: Strengthen the means of implementation and revitalize the global partnership for sustainable development.**

The project activates collaborations and partnerships with local or international public entities, especially to ensure technological development fair trade, and market access for cookstove production and distribution. Moreover, the mechanism of the voluntary carbon market itself allows mobilization of additional financial resources for developing countries, promoting the development, transfer, and diffusion of technologies and enhancing North-South international cooperation, as in the case of this project.

2.1.2 Description of the Project Activity

The clean cooking project developed by Eni S.p.A. and Eni Natural Energies - Rwanda branch (both referred as “Eni”) is a voluntary Grouped Project (GP) consisting of the distribution of Improved Cookstoves among local communities in the southern area of Congo, where the majority of households still rely on inefficient and pollutant traditional cooking techniques.

The activities in the field are performed by Eni together with a Project Instance (PI) actor called Rwanda Women Network (referred as “RWN”). RWN is a private and non-profit association with the aim of education, solidarity, social promotion and community development with focus on gender empowerment.

RWN operate in the country since 1997, establishing safe spaces across Rwanda. Initially safe spaces were built to support women who were victims of gender-based violence during the genocide, often infected with HIV, and simply had no place to seek help. Currently Rwandan women have access to 17 safe spaces across 7 districts. RWN also provide healthcare to over 29.000 community members in our owned and operated medical clinic in Kigali.

Moreover, RWN operate in the field of socio – economic transformation, providing financial education and invest in the development of women led local small businesses.

Within this project framework, RWN oversees communities’ engagement, improved cookstove distribution (storage, management, and household training) and following up with recipient households to assist them in the use of the device.

1. *Distribution of improved cookstoves*

The primary activity under this grouped project is the distribution of the improved cookstoves. The project aims to replace the traditional cooking systems currently used by local people (i.e. three-stone fires or rudimentary stoves) with improved cookstoves: these devices require less fuel and thus decrease the biomass combustion emissions, with a consequent reduction of the health hazards coming from the indoor smoke pollution. The introduction of more efficient stoves will help limit the demand for woody biomass, reducing the unsustainable use of forests and halting deforestation, which is highly increasing in the Country. The improved cookstoves' deployment will also have a positive impact on women's and children's lives, reducing the time they spend collecting wood and cooking meals, and giving them the possibility to invest their time in other activities (getting a revenue source or studying).

The distribution of improved cookstoves is not mandated by Rwandan law and the project is a voluntary initiative.

The cookstoves distributed are nowadays imported from an international producer. In the meanwhile, a cookstoves producers scouting is ongoing to identify reliable local producers with the necessary capabilities to fulfil the technical requirements. This will generate a strong impact on local content and promoting employment in cookstoves production as well as the creation of a robust cookstoves value chain once the scouting phase would be done. Once the local production will be able to produce the improved cookstoves quantity requested, all the ICS will be produced only by local producers.

The improved cookstoves distributed have higher thermal efficiency, compared to traditional cooking system, allowing for significant biomass fuel saving and a consequent reduction in GHG and smoke emissions, compared to the baseline situation. The current baseline scenario is characterized by the usage of inefficient cooking techniques with poor ventilation and high heat loss. To achieve the same thermal energy, according to the Clean Cooking Alliance, traditional cooking methods could require more than twice the amount of wood fuel compared to the improved cookstoves utilization.

According to the project framework, the improved cookstoves will be distributed to those households that prior to project implementation, as confirmed during baseline data collection, were relying on the use of three-stone fires powered by firewood or traditional bracers powered by charcoal, with no improved combustion air supply or chimney, for their cooking activities. Each household will receive an improved cookstove according to the type of woody biomass currently used as fuel for cooking activities to avoid creating difficulties for fuel supply.

2. Community engagement and training

The Project proponent, with the support of RWN, will carry out an important engagement and communication activity to inform, encourage, and promote cooking behavioural change at the household level. Brochures will be prepared and distributed to local communities to incentivize behavioural change sharing the benefits of improved cookstoves adoption. Awareness-raising activities at the village level will be carried out by RWN, followed by one-on-one training sessions with the single households during dedicated visits.

The project will also rely on communication channels at the local level (e.g. radio) to promote the project, inform the population, and guarantee project information dissemination. RWN will also

conduct follow-up visits in the villages to keep a regular exchange with the communities and support their transition to cleaner cooking technologies over time.

When the household receives the improved cookstoves, a dedicated training session is conducted to explain to the family members how the device works and how it has to be used and maintained. All households receiving improved cookstoves will be made aware of the environmental benefits that come from the use of these devices and the associated carbon credits that can be generated. The people voluntarily participate in the project, signing an End User Agreement (“EUA”) with Eni confirming that any emission reduction resulting from the use of the improved cookstoves delivered is owned by Eni. The personal information of the beneficiary household and the serial number of the improved cookstoves are noted in the EUA and stored in a data management system which allows Eni to uniquely identify each device by tracking the GPS coordinates of the beneficiary household and the improved cookstove delivery date. The project proponent is going to collect the personal data of the designated household member, taking notes of the house’s GPS location and the unique identification number of the delivered stove. According to the VCS methodology VMR0006 v. 1.2 for the project aimed to be labelled by SD VISta, those data are going to build up a database of the people necessary for the monitoring and verification activities.

Beneficiary families are advised about the visits by the Implementer and the VVB during the meeting for stakeholder engagement and when the Improved Cookstoves are delivered, Additionally, if they are chosen for the visit, they will be notified as soon as possible before the VVB arrives, either directly via the implementer field staff or by their cellphone number, if they have given one.

2.1.3 Implementation Schedule

Date	Milestone(s) in the Project's Development and Implementation
14 July 2023	Final report on-field baseline activities
16 November 2023	Project Internal authorization
17 November 2023	Signature of Project Implementation Agreement with Implementing Partner
5 December 2023	First stakeholder consultation meeting
14 December 2023	First batch of improved cookstoves delivered

2.1.4 Project Proponent

Organization name	<i>Eni Natural Energies S.p.A. – Rwanda Branch</i>
--------------------------	--

Contact person	<i>Martufi Mauro</i>
Role in the Project	<i>Branch Manager</i>
Address	<i>Blue Star House, 35 KG 7 Ave, Kigali, Ruanda</i>
Email	<u>mauro.martufi@eni.com</u>
Organization name	<i>Eni S.p.A.</i>
Contact person	<i>Matteo Pozzani</i>
Title	<i>Carbon Offset Technologies Specialist</i>
Address	<i>Via Emilia 1, San Donato Milanese (MI), 20097, Italy</i>
Email	<i>matteo.pozzani@eni.com</i>

2.1.5 Other Entities Involved in the Project

For the field activities implementation, Eni has involved Rwanda Women's Network (RWN), a Rwandan Non-Governmental Organization. RWN supports Eni as a supplier for the community's engagement activities, improved cookstove distribution and follow-up with beneficiary households.

In the implementation agreement signed between Eni and RWN, a clause regarding the carbon credits rights has been included saying that Eni is the owner of carbon credits coming from the project activities implementation and RWN does not have any kind of claim on these.

Organization Name	Rwandan NGO Rwanda Women's Network
Role in the Project	Responsible for ICS distribution and monitoring activities
Contact Person	Mary Balikungeri
Title	Director and Founder
Address	KG 504, Kagugu, Kigali, Rwanda
Telephone	+250 788 334 257
Email	info@rwandawomennetwork.org

2.1.6 Project Type

The Project is a non-AFOLU Grouped project. The general sector is energy efficiency.

SD VISta sectoral scope	SDG
1. Agriculture, forestry and other land use	15. Life on Land
2. Climate Change Adaptation	13. Climate action
4. Energy	7. Affordable and Clean energy
7. Health	3. Good Health and well being
10. Livelihoods	1. No poverty 8. Decent Work 10. Reduced inequalities
14. Women's empowerment	5. Gender equality

Table 2: SD VISta sectoral scope and SDGs

2.1.7 Project Location

The project is deployed in Eastern Rwanda area, considering the districts listed in Table 3. PI1 started in Nyagatare district.

The Nyagatare City's geographical coordinates are the following:

- 1°18'52.38" S
- 30°22'20.71" E

Table 3: Project areas

No	Project Province	District
1	Eastern Province of Rwanda	Nyagatare, Gatsibo, Kayanza, Rwamagana, Bugesera, Ngoma, Kirehe

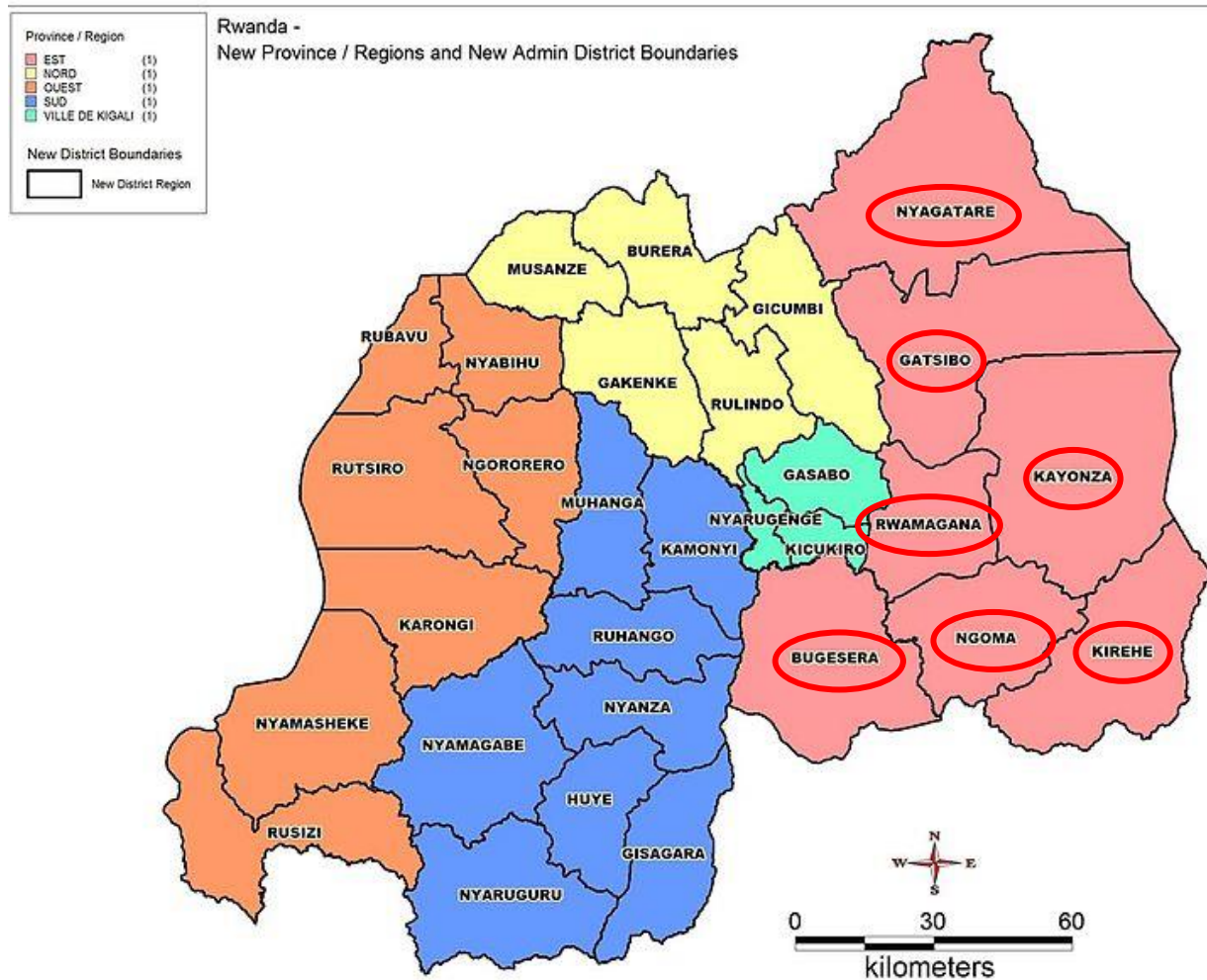


Figure 1: Geographical map of Rwanda with province representation

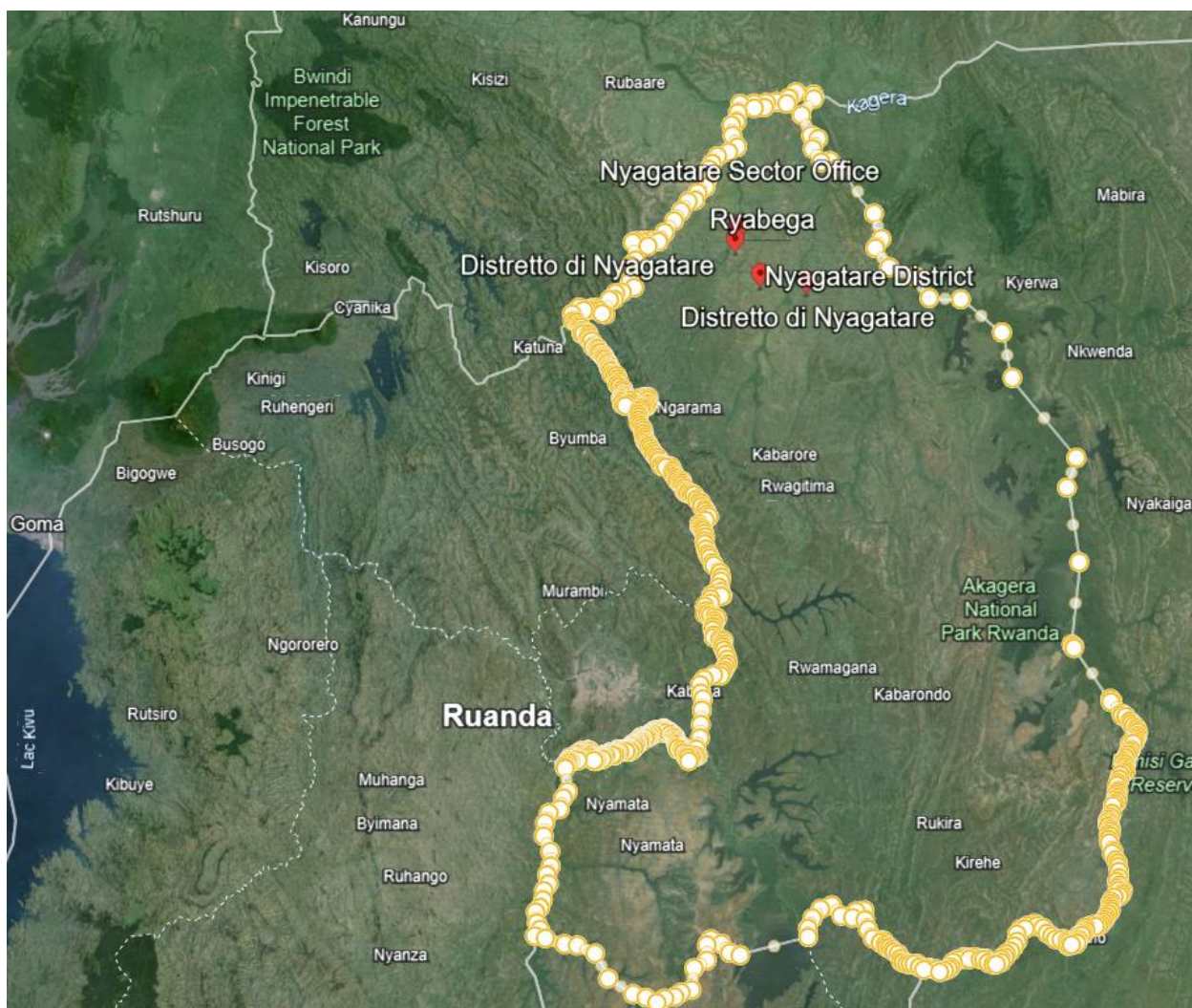


Figure 2: Orthophoto of Rwanda with the project area identification (source: Google Earth)

2.1.8 Baseline Scenario

Social economic conditions:

Rwanda is a poor country with more than 50% of population living in poverty conditions. Rwanda is predominantly populated in rural areas; indeed in 2022, only 18% of population live in urban areas, while 82% of the population live in rural areas, according to World Bank.

Despite the GDP of the country is increased in the last decades, the economic growth does not benefit the entire population, but for the most part it benefits the increasingly large middle class that lives in the capital. Most inhabitants of rural areas still survive with great difficulty.

More than 50% of the population lives below the poverty threshold. In fact, on average a Rwandan citizen survives on less than \$1.50 a day.

Despite recent attempts to improve infrastructure, the economy is still based on the export of products from tea and coffee plantations, where most of the country's workforce is employed, while industry is still underdeveloped.

According to the World Bank database, in Rwanda in 2021, only 5% of population have access to clean fuel and technologies for cooking. Firewood and charcoal are the main energy sources especially in rural areas; women and children, especially girls, are usually responsible for the collection and cooking activities of the household.

National capital conditions

Rwanda has a diversified natural ecosystem ranging from afro-montane in the Northern and Western Provinces to lowland forests, savannah woodlands, savannah grasslands in the Southern and Eastern Provinces. Rwanda is also rich in large numbers of inland freshwater lake and wetland ecosystems. These ecosystems provide a wide range of goods and services, but they are under pressure from human activities. The most critical ecosystems include forests, water, wetland ecosystems, among others. Some of these have been already converted into agricultural use and settlements.[36]

Rwandan forest area was estimated at 240,746 ha in 2007. This translates to approximately 10 per cent of the surface of the national dry lands (23,835 km²).

Statistics from the Forest department show that forests were estimated to cover 659,000 ha in 1960 (ROR 2004). This reflects a loss of approximately 64% of forests in between 1960 and 2007, which is more than 1.3% per cent per year. The rapid increase in population is increasing pressure on forests in terms of encroachment and deforestation.

Moreover, from 2001 to 2023, Rwanda lost 46.2 kha of tree cover, equivalent to a 9.3% decrease in tree cover since 2000, and 32.6 Mt of CO₂e emissions, as shown in figure 3.

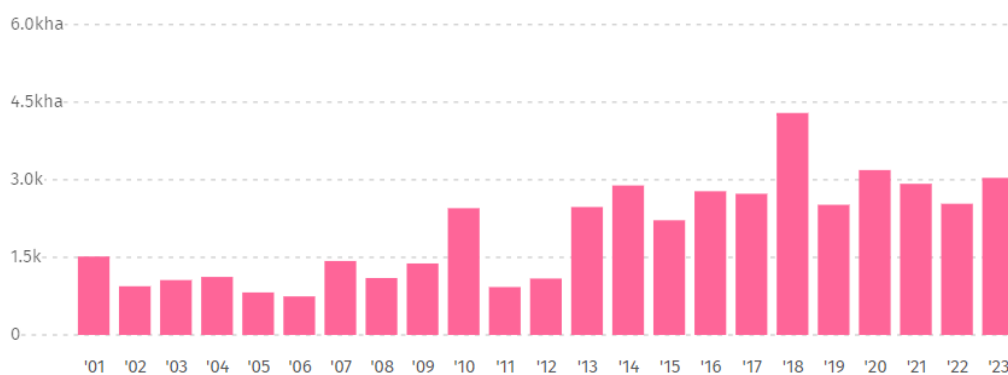


Figure 3: tree cover loss in Rwanda (kha/year) (source: Global Forest Watch)

In Rwanda from 2001 to 2023, 12% of tree cover loss occurred in areas where the dominant drivers of loss resulted in deforestation. In figure 4 are shown the main deforestation drivers at 2023.

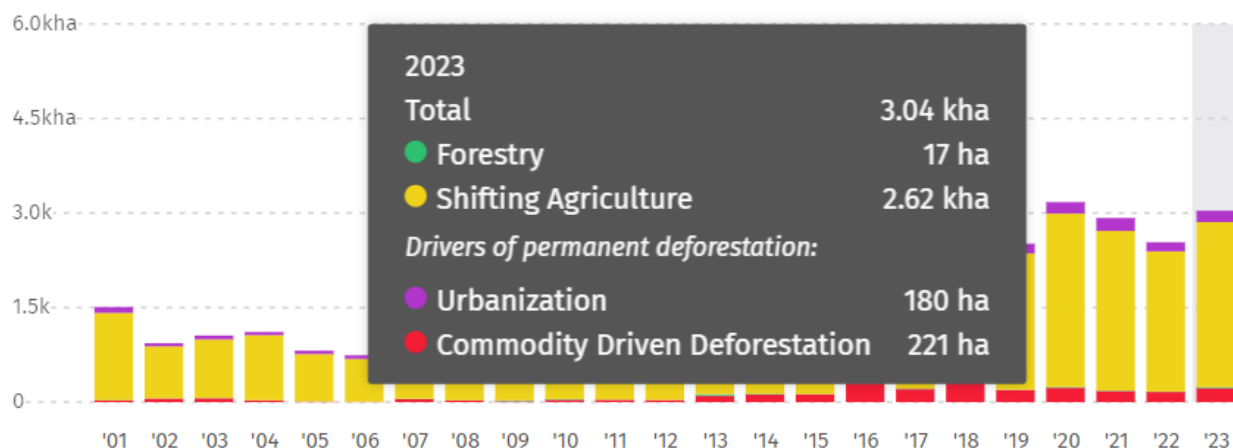


Figure 4: main deforestation drivers from 2001 to 2023 in Rwanda (source: Global Watch Forest)

In addition, Rwanda is vulnerable to natural disasters due to its geographical location (East African Rift Valley, near Nyiragongo volcanoes) and the effect of the Inter Tropical Convergence Zone (ITCZ) on the continent. Such disasters are exacerbated by poor farming practices, deforestation and environmental degradation among others.[37] [38]

2.1.9 Causal Chain(s)

The causal chain tree describes the effects of the project activities on several SDGs. Both the indicators impacted directly (solid line) and those impacted indirectly (dotted line) are indicated, consistently with what is reported in the first section of this project document.

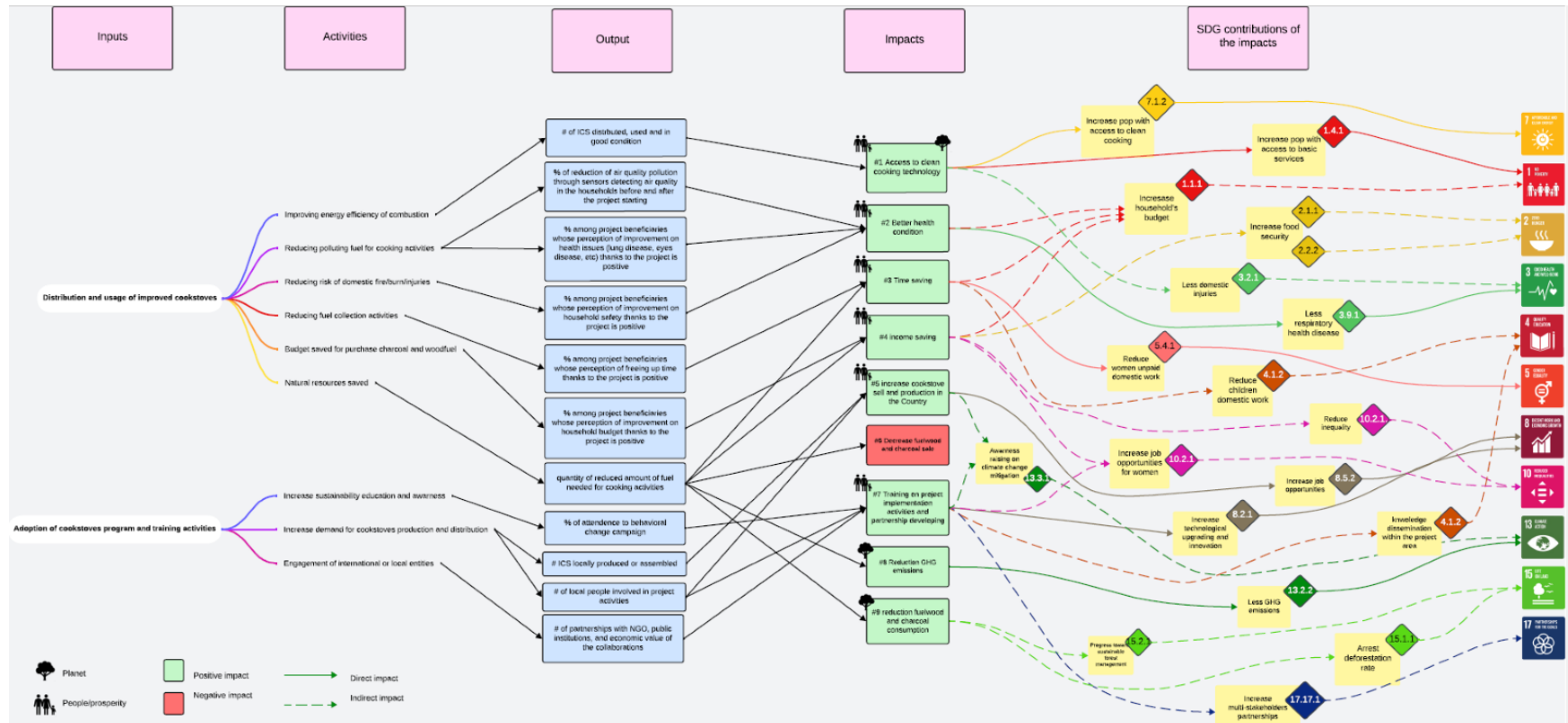


Figure 5: Project Value Chain

2.1.10 Threats to the Project

- Local acceptance: any conflict with the local community could delay or stop the project.

Solution: Identifying, analysing and listening to stakeholders are fundamental elements for Eni: knowledge of expectations and transparency of relationships are the basis of the commitment to build trusting and lasting relationships with its own stakeholders in the areas where Eni is present.

Eni has implemented a specific web-based platform called Stakeholder Management System (SMS) to carry out this daily and proactive engagement with stakeholders at local, national and international level. This tool makes it possible to:

- map, classify and prioritise stakeholders and their requests regarding sustainability issues;
- monitor the stakeholders involved in each project/area of presence, storing every interaction with them and analysing their requests;
- highlight any critical issues and areas for improvement for the stakeholder engagement strategy to be developed (i.e. potential impacts on human rights, possible presence of vulnerable groups).

2.1.11 Benefit Permanence

To maintain the benefit permanence of the project, the following activities are planned:

1. To associate the project under the GHG mechanism, i.e. VCS and SD VISta, to ensure that it is operational and yields carbon credits at least throughout the project lifetime. This includes also all the monitoring activities to ensure high quality carbon credit for the entire lifetime of the project: check on end users, support assistance, and maintenance interventions.
2. To promote local manufacture. Although in the early phase of the project the devices distributed will be imported from international manufacturers, in the meanwhile a local production chain will be implemented in order to enhance the local content and promote employment in the cookstoves' industry. For the project purpose, it is fundamental that the issued improved cookstoves have a thermal efficiency higher than 25% certified by the producer, therefore not all cookstoves are eligible for the project implementation and this target requires experience and proper machinery. In the long term, Eni has the goal to help the local manufacturer to increase its improved cookstove volume production, increasing also the local employment in the supply chain in the Country. By doing this, one of the goals of the projects in the next years is the transfer of technology to local craftsmen. Collaborating with craftsmen already skilled on this topic, providing training and courses (regarding both technical and project management aspects) could be put in place, to give the proper knowledge to identify a reliable material supply

chain and manufacture improved cookstoves compliant with VERRA's requirements of methodology VMR0006.

3. To offer a complete and long-term preparation and documentation to the households involved in the project. This includes an important awareness campaign to support behavioural change at the household level. Brochures will be prepared and distributed and engagement activities for financial literacy and environmental and health benefits awareness will be shared with all the local communities involved in the project.

The awareness raising activities are performed during the cookstoves distribution and continues along the project's lifetime. Those are designed to make the cookstoves recipients aware of all the environmental and socio-economic benefits which are expected to be maintained and enhanced.

Moreover, through awareness raising activities, the project aims to establish a direct contact between people and local producer, allowing for a responsible cookstove management and putting the basis for households to purchase cookstoves directly by their own in the future.

2.2 Stakeholder Engagement

2.2.1 Stakeholder Identification

As requested by VCS Standard, Eni has conducted some local stakeholder consultations with RWN's support before the project's start, to adequately inform the local communities in the area regarding the main pillars of the project and the implementation of the activities. The consultations also allowed the stakeholders to express their points of view regarding the project and ask questions and clarifications; all this information has been integrated into the project design for improvements.

The stakeholders invited to take part in public consultations belong to different areas. According to UNDP stakeholder engagement guidelines (2022) [16], broad stakeholder categories, beyond the project board, are split into direct affected groups and indirectly affected groups with interest or influence in the project (Figure 6).

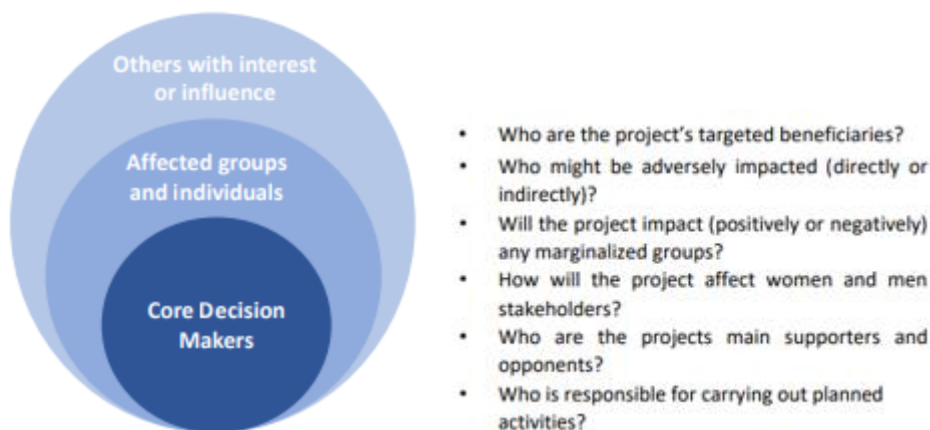


Figure 6: broad stakeholder categories; Figure 7: key questions; Source: UNDP [16]

The identification of relevant stakeholders follows the UNDP guidelines that are summarized in key questions like the ones in Figure 7. The process identification has been conducted following each step of the project implementation: from the project design to the cookstove distribution, both categories of direct and indirect groups have been identified.

1. "Project affected parties": individuals, groups or entities that are affected or likely to be affected by the project:
 - a) current people;
 - b) General public and local people interested in the clean cooking project as potential people.
2. "Other interested parties": individuals, groups or entities that may have an interest in the project:
 - c) Representatives of relevant Ministries from central and regional offices

- d) Political representatives and administrative authorities at regional and local levels
- e) NGOs and member of youth and women organizations;
- f) Scholars, teachers and academic institutions;
- g) Representatives of industrial sectors and manufacturers involved in the cookstoves supply chain in the project area;
- h) Media (e.g. radio)

2.2.2 Stakeholder Description

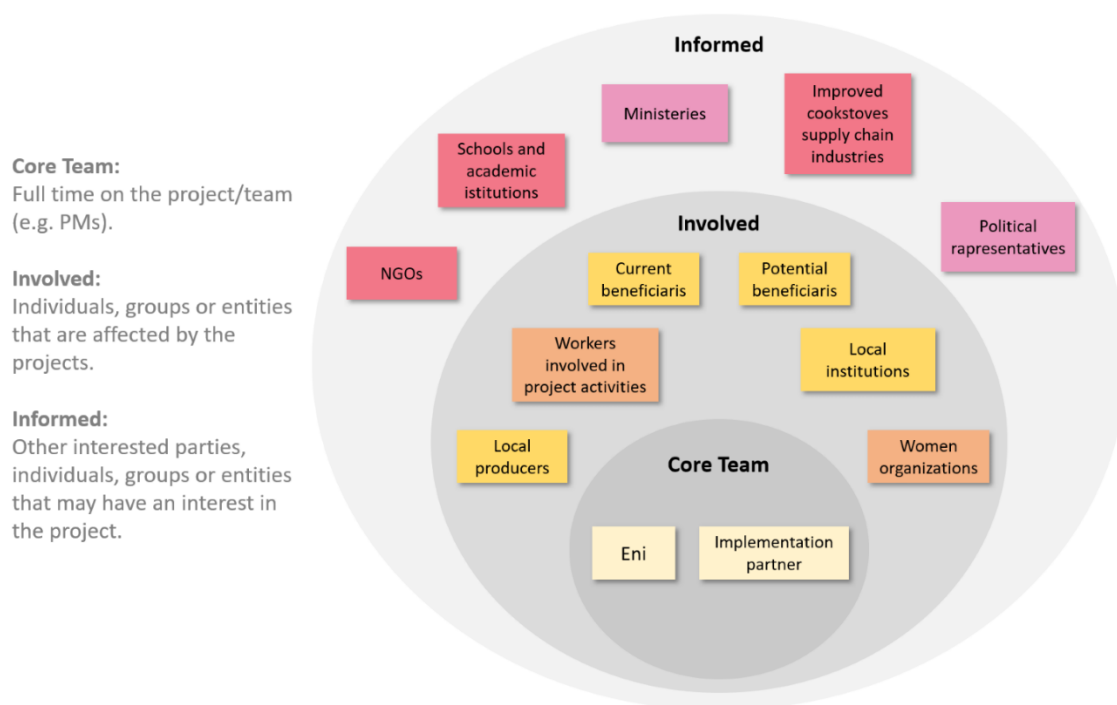


Figure 8: Stakeholders identified

Following the identification process showed in the previous section and summarized in fig 9, the identified stakeholders can be defined as:

1. Project affected parties.
 - a. current people: household members are identified as individuals who are directly influenced by project activity, especially women, as they are usually in charge of cooking preparation and activities in their families. The project does not discriminate against any category of marginalized individuals (elderly, disabled, etc) and does not inflict any adverse impact on these categories. All the villages in the project area are eligible, with no hierarchy to establish any priority to receive the improved cookstoves, unless for those households that already use gas or electricity as a fuel (cleaner technologies) and to whom improved cookstoves will not be distributed since they are

not eligible for carbon credits generations according to verra methodology VMR0006 v. 1.2.

- b. Local people interested in the clean cooking project as potential ICS people: through the active communication campaign, interaction with communities, and stakeholder consultations, new people can be reached. The potential people typically share similar characteristics with the current ones: no village or household will be excluded a priori from the improved cookstove distribution, and the only requirements are related to the current cooking methods and the involvement in other clean cooking projects, in compliance with the VCS verra methodology VMR0006 v. 1.2.
- c. Local community leaders: chiefs of the villages in the target area of the project. Community leaders are directly involved in project activities prior to the distribution activities. In particular, in the phases of the village households census, community engagement and awareness on the topics related to the project, and collective explanations about the activities that will be carried out in the village.

2. Other interested parties:

- a. Representatives of Ministries involved in the project topic: the identified ministries are: Mayor of Nyagatare District; European Union Ambassador in Rwanda; Minister of Infrastructure; Minister of State; Permanent Secretary; CEO of Rwanda Development Board (RDB); Director General of REMA, national and international NGO representative.
- b. Ministries representatives received an email or a letter with the formal invitation to the meeting held close to their head office location, with, in attachment, a summary of the clean cooking project's key points.
- c. Political representatives at local levels: District Secretaries; Block Chiefs; Community Leaders. Similarly, to ministries representatives, local authorities received formal invitation to meetings.
- d. NGOs: any organization that operates in the same geographical areas and acts on similar projects. Moreover, possible collaboration and synergies with other local NGOs on clean cooking projects have been investigated during stakeholder consultations.
- e. Media: Any national or international media interested in spreading the project's information. It is not excluded that the project can also rely on communication channels at the local level (i.e. radio) to promote the use of improved cookstoves and to maintain a close relationship with the population.
- f. Supply chain industries: any representative of industrial sectors and craftsmen involved in improved cookstoves' supply chain located in the project area. Indeed, identifying craftsmen already skilled on this topic or interested in training and courses (regarding both technical and project management aspects) allows to give the proper knowledge to a reliable material supply chain and manufacture improved cookstoves (unless is compliant with VERRA's requirements).
- g. School and academic institutions: education institutions, playing also central role in villages' and cities' cultural life.

2.2.3 Stakeholder Consultation

The stakeholder consultations have been carried out between in December 2023 through public meeting held inside the project boundaries.

Eni has conducted local stakeholder consultations with RWN to adequately inform the local communities in the area on the main pillars of GP implementation. The consultations also enabled the stakeholders to express their point of view on the project and ask questions and clarifications; all this information has been integrated in the project design for improvements.

The stakeholder consultations have been carried on December 5th 2023, in Nyagatare City.

The dates and locations of the public meetings held are:

- Nyagatare, 5/12/2023, Moon's Charity Garden

For the invitations, different approaches were adopted to ensure that each stakeholder group was adequately reached and informed on the meetings' time, location, and subjects.

Ministries and local authorities received a letter with a formal invitation to attend the meeting, with attached a description of the project.

People interested in project activities on the ground, such as NGO members, women's associations, social workers, and local households were invited through posters attached in the villages and cities close to the meeting location at least two weeks before the day of the consultation. The content of the posters was like the description shared with the political authorities, explaining the project's goals and implementing guidelines, such as:

- A brief introduction about Eni company and its decarbonization strategy, describing in simple terms the nature of carbon credits and what is their role to achieve the decarbonization.
- The purpose of the clean cooking project (namely offsetting the Scope 1 and Scope 2 GHG emissions coming from oil and gas field production), an introduction to the Eni presence in the country and the new hydrocarbons discovery made by Eni offshore. It has been stated very clearly that carbon credits generated through the clean cooking project are not going to be traded on the international market but used to achieve internal offsetting targets.
- The main features of the project (lifetime, area involved, cookstove products, type of beneficiary households involved, advantages coming from devices' adoption instead of relying on inefficient techniques), describing in detail the distribution mechanism and the project additionality. The improved cookstoves will be deployed without any upfront cash payment requested to the local people, and through an agreement with Eni the households waive any rights on the carbon credits ownership coming from the emissions reduction.

All these points were thoroughly presented and discussed in detail during each consultation, giving the possibility to the stakeholders to ask clarifications or expose their

grievances/opinions/suggestions. To maximize the feedback received from the population and to allow everyone to make comments or express concerns about the project, a dedicated email address was created and contact information was shared during the consultations (info.cookstove.rwanda@eni.com).

Before the conclusion of the meeting, an attendance sheet was circulated and signed by the attendees (those with writing ability and willing to share their information), including personal data for their identification.

Stakeholder Meeting in Nyagatre City

Date of stakeholder consultation	<i>5th-December-2023</i>
Stakeholder engagement process	<i>The stakeholders have been engaged via mail or with public announcement and posters in the area around the meeting location. The communication has been done in English.</i>
Consultation outcome	<i>The main outcomes of the consultation have been the ICS distribution mechanism, the project additionality (ICSs deployed without any upfront cash payment), the monitoring activities and the positive impacts on local communities and environment</i>
Ongoing communication	<i>Stakeholders could communicate through a dedicated mailbox (info.cookstove.rwanda@eni.com) or by phone calling with focal point appointed in each village of cookstoves distribution</i>
Stakeholder input	<i>Pastor Jacque for the Restoration Church at Nyagatare, appreciated the advantages of improving cooking stoves project. That this will help their church members to be in the fellowship on time.</i> <i>Q: Will that stoves be used from anywhere?</i> <i>A: The stoves are not allowed to be used from anywhere, but on plain surface.</i> <i>Mr. Kazayire from Burumba, noted that once he receives the stove, he will be helping his wife to cook this time around and gender disparity is going to be addressed in Nyagatare.</i> <i>Mr. Bahati Paul from Nyagatare town, cut less trees and plant more. This project is going to enable us cut less trees because we shall be using less firewood.</i> <i>Mary Claire from CNF: Appreciated the expected impact of the use of improved cooking stoves that will promote gender-equality. She noted that women will have enough time to concentrate on other income generation activities due to the reduction on time spent in the kitchen.</i>

Q: is the stove smaller than his cooking saucepan?

A: This kind of stove has ability to serve for any sauce pans families use, but not for larger sauce pans that are used in hotels.

Mukamusoni Jane from Barija hailed the provision of improved cooking stoves underlining that her eyes were affected by smoke because of using old traditional cooking techniques.



2.2.4 Continued Consultation and Adaptive Management

During the consultation meeting, the participants were informed about how to keep in contact with the project proponents, either through direct contacts (phone /email) or through representatives of different groups. RWN is implementing a communication campaign at all levels (prefecture, sous-prefecture, villages, and households) and stakeholder inputs and feedback, especially from academic institutions, users, and potential users, are encouraged to ensure continuous improvements. Specific adaptive management, according to households' needs, will be made, especially in the one-by-one training session with the single household during the dedicated visits by RWN. In general, prior notice will always be granted giving the possibility for interested people to be available in the village during the distribution activities.

2.2.5 Anti-Discrimination

The distribution of improved cookstoves to local communities using traditional cooking technology occurs without any distinction in gender, race, religion, educational backgrounds, or any other aspect. Eni fosters an environment of mutual respect where employees and business partners can work together without fear of harassment, discrimination, or infringement on human rights. Eni has strict policies against discrimination and in support of diversity and inclusion and any program implemented by Eni and its implementing partners ensures that all applicable laws, regulations, and international standards, including the Universal Declaration of Human Rights and core ILO Conventions, are adhered to. The United Nations Guiding Principles on Business

and Human Rights (UNGPs) and International Conventions are key references for Eni for guiding activities, regulatory system, and governance towards the respect for and promotion of human rights. Eni's commitment is reflected in Eni's Declaration on Respect for Human Rights and in a series of documents relating to Eni internal organisation and relations with external parties, such as the Code of Ethics, the Sustainability Policy and the Supplier Code of Conduct. Eni's approach to human rights is also confirmed in the international Global Framework Agreement on Industrial Relations and Corporate Social Responsibility, which was renewed in June 2019, and has an explicit reference to the UNGPs. Eni is also committed to the Voluntary Principles on Security and Human Rights (VPSHRs) and it was accepted as an “Engaged Corporate Participant” in the Voluntary Principles Initiative (VPI), the multi-stakeholder initiative involving governments, corporations and NGOs. In 2021, Eni also prepared an Annual Report describing the activities carried out during the year to implement the VPSHRs, a set of principles to support companies regarding respect for human rights in the management of security operations, defined and promoted by the VPI. Finally, Eni ranked first in the Corporate Human Rights Benchmark (CHRB) Human Rights Index in the latest study carried out in 2020, equally with just one other company.

2.2.6 Worker Training

One of the project's targets is to maximize local cookstove supply with a direct contribution to job opportunity creation. For the time being, the improved cookstoves are only purchased from international manufacturers satisfying the requirements. In the meanwhile, the project will support the local producers to establish and increase the improved cookstoves volume production and expand its operations, setting up regional hubs promoting local employment. The project will support the firm to increase the improved cookstove volume production and expand its operations, setting up regional hubs promoting local employment. Local craftsmen will be involved in the technology transfer through training and courses. Besides technical skills, the workers will be trained about the management structure, regulations, and worker rights.

Moreover, the necessary monitoring plan for the project activities will be carried out by a monitoring team. The team will be comprised of local people speaking the local languages and fully aware of local traditions and habits, also to increase local participation in project implementation.

2.2.7 Equal Work Opportunities

The work opportunities within the project framework are promoted following criteria of equal opportunities, diversity, and inclusion. The individuals for the monitoring plan will be selected based on their previous competence and experience and will be trained accordingly for each monitoring activity. Recruitment advertising will be given before the hiring for the project and will be provided through written advertising and proper oral announcement by communication campaign to make sure that all stakeholders, no matter their gender, race, religion, will be given an equal opportunity to fill all work positions.

Eni approach has always been rooted in the international culture of plurality. It is based on the fundamental principles of non-discrimination, equal opportunity, and inclusion in respect of all forms of diversity, as well as integrating and balancing work with personal and family needs. Eni has chosen to discuss these issues with both national and international peers by joining organisations and participating in inter-company initiatives and benchmarking systems for sharing best practices and identifying common activities in Diversity & Inclusion. They are the European Round Table (ERT), World Economic Forum (WEF), Industrial ALL Global Unions and the UN Global Compact, among others. Eni has also signed up to the Women Empowerment Principles, the gender equality and women's empowerment initiative of UN Women and the UN Global Compact.

2.2.8 Workers' Rights

The project activity includes production and distribution of improved cookstoves and monitoring plan activities which involves local manufactures and workers. Eni is committed to respect and promote internationally recognized workers' rights in all the Countries where it operates. This commitment goes beyond the International Bill of Human Rights and the International Labour Organization (ILO) Declaration on Fundamental Principles and Rights at Work, which together constitute the minimum requirements of businesses according to the UN Guiding Principles, and includes also:

- the development of a comprehensive framework of internal policies and rules to affirm the commitment to respecting internationally recognized workers' rights that could be most relevant to Eni's business.
- The commitment of Business Partners and suppliers to also comply with them.
- The collaboration with unions and the engagement with international organizations including the ILO.

In April 2020, Eni published the Supplier Code of Conduct: it establishes the mutual commitment to recognize and protect the value of all the people, commitment to contrasting climate change and their effects, operating with integrity, protecting company resources, promoting the adoption of these principles within their own people and supply chain. Among these principles, regarding human rights and work, the reference is the International Labor Organization (ILO) Declaration on Fundamental Principles and Rights at Work. All suppliers will be asked to sign the Code and to promote the principles contained in it along their supply chain. Specific commitments on human rights are:

- Prohibit the forced labor, the undeclared labor, the compulsory labor and all the forms of modern slavery and human trafficking.
- Prevent any form of work by children under the age of 15 and ensure, in compliance with local law, that teenagers under the age of 18 are not employed in hazardous jobs.

- Abide by working times and rest periods in compliance with the applicable legislation and in line with international standards.
- Prevent any kind of discrimination.
- Ensure respect of workers' rights and trade unions freedoms.
- Establish clear and fair working conditions defined in the employment contract.
- Respect the cultural, economic, and social rights of the local communities, minorities, indigenous peoples, and other vulnerable groups.

In line with Eni corporate culture, which promotes inclusion and respect for uniqueness, in December 2021 Eni issued a policy against violence and harassment in the workplace. The Zero Tolerance policy, in accordance with the principles of the Geneva Convention of the ILO, defines a broad perimeter of types of harassment that allows to identify misconduct and behaviour that should not be engaged in and should be reported.

2.2.9 Occupational Safety Assessment

The improved cookstoves are produced with safe materials (mainly made of stainless steel), that do not release any toxic substance and do not represent any kind of threat to human health or the environment. This is valid both for the workers in the production and distribution chain and for the end users. Also, the improved cookstoves are safer than traditional cooking systems: for instance, during the combustion, the handles are not as hot as in the case of traditional stoves, and the combustion is limited in an enclosed space thus reducing the risk of burning accidents. Due to the improved technology, the cookstove design is more suitable to reduce heat dispersion and to avoid blazing flame outside the device. Thus, the improved cookstoves do not bring any safety issues: they reduce safety hazards. However, cooking activities shall always be performed by adults as with any kind of cooking technology since they are not made for children.

2.2.10 Feedback and Grievance Redress Procedure

For the management of Carbon Offset project, Eni has developed a grievance process providing guidelines and principles to be followed also by the third-party implementing partner developing a clean cooking project.

The Grievance Mechanism:

- is provided at no cost to the communities involved in Project activity;
- provides a recourse mechanism that could involve third parties in case Eni /Implementing Partner and the complainant do not reach or agree upon a resolution – the identified must meet the requirements of good reputation and professionalism, also from an ethical and moral point of view, ensuring impartiality in the process of assessment and proposed resolution;

- does not impede access to judicial or administrative remedies;
- allows for anonymous grievances;
- allows lodging from both collective and individual grievances.

During the consultation, the stakeholders had the possibility to ask for clarifications and expose grievances/opinions/suggestions. A dedicated phone number and email address were shared in case further information was needed or in case of an issue to be solved. All this information is shared also through the dedicated communication campaign, through brochures, posters, and social media channels.

The feedback and grievance procedure ensures that every issue is properly prioritized and addressed.

In Figure 9 below there is a schematic description of the management of grievances.

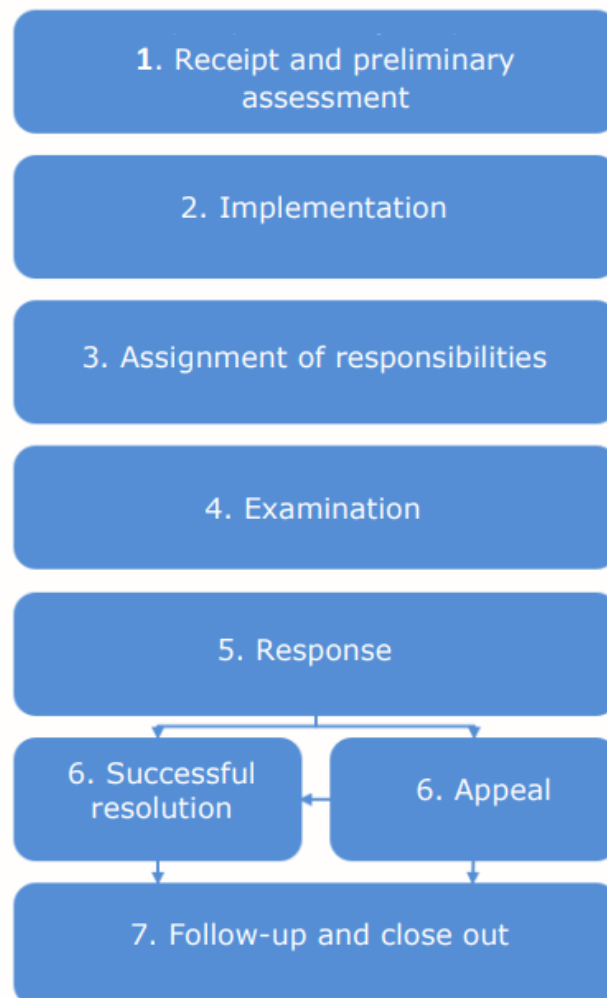


Figure 9: Management of the grievance process

As a first step, the procedure involves a preliminary assessment, that includes a preliminary classification and prioritization of the grievance. The person (or group of persons) who expressed a grievance is promptly notified that it has taken charge of it. Together with the acknowledgment of receipt of the complaint, it is indicated also, where possible and appropriate, the activities planned for the management of the grievance and the timeframe that can reasonably be expected for the conclusion of the process. The preliminary assessment also involves the assignment of responsibilities and verification to the proper functional units to examine the grievance. In parallel with the ongoing examination procedure, the complainant is kept adequately informed about the progress of the procedure.

Once a proposed resolution has been approved, the response is notified and discussed with the complainant, who may agree (successful resolution) or disagree. In the latter case, the complainant's response is examined again and it may be possible the involvement of a third party to resolve the grievance.

2.2.11 Feedback and Grievance Redress Procedure Accessibility

During the consultation, the stakeholders had the possibility to ask for clarifications and expose grievances/opinions/suggestions. A dedicated email address were shared in case further information is needed or in case of an issue to be solved and are always operated through the lifetime of the process. Moreover, through the engagement and monitoring process, the people are asked to provide feedback and there is the possibility to assess grievances. All this information is also shared through the dedicated communication campaign, through brochures, posters, and social media channels.

2.2.12 Stakeholder Access to Project Documentation

During stakeholder consultations, all the project details were shared among participants. Considering that many project documents are very technical, a comprehensible non-technical summary detailing the project activities and outcomes was also distributed after the stakeholder meetings. Posters with the summary description are also attached in the villages and cities. Verbal information is also provided during the distribution phase, together with phone and email contact to reach field staff implementors for further clarification. All the project details are available both at the centralized level and at the local office in the host country.

The full project documentation will be uploaded on Verra website.

2.2.13 Information to Stakeholders on Assessment Process

People are informed to be part of the project that distributes improved cookstoves free of charge with benefits for health, safety, and the environment. Households are also aware in advance that the use of improved cookstoves generates carbon credits, and, during the distribution of the devices, an agreement is signed between the households involved and Eni: each household receives the device without cash uptake in exchange for the environmental benefits' property

resulting from cookstove use. Project implementors also inform the users about the necessity of an ongoing monitoring process to track the outputs over and above phase. During stakeholder meetings, it has been stressed that every household in the project area is free not to participate in the clean cooking project. Although the improved cookstove is given free of uptake cash payment, no one is forced to accept it: project membership is voluntary, and the device will be distributed only to those households that are interested in using the new technology.

The delivery date and site visit for each village will be shared in advance and will be set in accordance with the village chief or board and women's organizations. Prior notice will be always granted giving the possibility for interested people to be available in the village given their working commitments.

The project proponent and implementor will use all the available communication channels at the local level (i.e. radio) to promote the project, inform the population, and guarantee project information dissemination. RWN will also conduct follow-up visits in the villages to keep a regular exchange with the communities and support their transition to cleaner cooking technologies over time.

All people of the project are informed of the carbon credit generation from the project implementation. They are aware that the validation and verification processes include site visits by VVBs accredited in the carbon credits generation register and when the visits occur they are informed in due time through the reference focal point of each village.

2.3 Project Management

2.3.1 Avoidance of Corruption

Eni considers corruption to be an intolerable obstacle to business efficiency and fair competition. In November 2009, to implement this commitment in the business, the company voluntarily extended its compliance program to the risks of corruption by adopting the Anti-Corruption Guidelines and their Ancillary Procedures. Furthermore, Eni's Whistleblowing reporting channel allows employees, members of corporate bodies or third parties to forward (including confidentially and/or anonymously) complaints relating to problems in the internal control system (compliance with law and corporate policies, guidelines or procedures etc.) or other violations of the Code of Ethics (issues involving ethical conduct, mobbing, harassment, conflicts relating to human resource management, etc.).

The Anti-Corruption Compliance Program is rooted in the existing national and supranational regulations concerning corruption and money laundering and, within Eni, is embodied, from a regulatory point of view, in the "Anti-Corruption Management System Guideline" as well as in other detailed regulatory instruments (so-called Anti-Corruption Regulatory Instruments) which constitute the reference framework for identifying activities at risk and the control instruments

that Eni makes available to its People to prevent and combat the risk of corruption and money laundering. The Compliance Program is mandatory for Eni and all its Subsidiaries, in Italy and abroad. Eni also recognises the importance of clearly communicating its values and ethical principles to its counterparties/partners, providing for adequate anti-corruption and anti-money laundering compliance commitments in the relevant contracts.

Eni also considers transparency a corporate value and views its promotion as crucial for a more inclusive resource governance in favour of communities' interest and as a prerequisite to fight international corruption. Eni's commitment is confirmed by active participation, since 2005, to the voluntary multi-stakeholder initiative of EITI (Extractive Industries Transparency Initiative) and by the support to its efforts in promoting an open and accountable management of natural resources.

2.3.2 Statutory and Customary Rights

The project takes place both in private property and in public areas. The implementor staff enters the private property of the household only with the explicit permission of the owner. As already stated, project membership is voluntary and the device will be distributed only to those households that are interested in using the new technology. Stakeholder meetings took place in public buildings and demonstration meetings at neighbourhood level are organized on public land, i.e. at a central village meeting point, or on private property with the explicit verbal invitation of the owner.

2.3.3 Recognition of Property Rights

The distribution of the improved cookstoves is free of charge for the people. To take part in the project, an agreement must be signed by the recipient to accept the cookstove and to waive any rights on the environmental benefit in favour of Eni. After this agreement, the recipient has the property of the cookstove. The implementor staff does install the device only at the property owner's invitation.

2.3.4 Free, Prior and Informed Consent

The implementing staff installs the cookstove only at the informed invitation of the household. The households are free to choose whether they want to be part of the project or not. Free, prior, and informed consent is obtained by signing the agreement before the distribution, which clarifies the cookstove property right of the user and the wave of any right for the generation of carbon credit in favour of Eni.

2.3.5 Restitution and/or Compensation for Affected Resources

The project activities do not affect any party's land and resource on their land.

2.3.6 Property Rights Removal/Relocation of Property Rights Holders

The project activities involve the distribution of improved cookstoves only and do not involve the removal or relocation of property rights holders from lands or territories, nor do they force rights holders to relocate activities.

2.3.7 Identification of Illegal Activities

There is no transfer of funds to the people. The personal information of the beneficiary household and the serial number of the improved cookstoves are noted in the End User Agreement and stored in a data management system which allows Eni to uniquely identify each device by tracking the GPS coordinates of the beneficiary household and the cookstove delivery date. This should avoid theft and corruption activities.

2.3.8 Ongoing Conflicts or Disputes

This issue is not applicable since the project activity involves distribution and installation of improved cookstoves to individual households only. These activities do not affect property rights or property disputes.

2.3.9 National and Local Laws and Regulations

Eni and RWN adhere to local and national laws of the Country. In the Country, there are no mandatory laws, policies, or regulations which would oblige households to use energy efficient cookstoves.

The literature review on this field includes:

- The Rwanda Energy Policy (REP, 2015)
- Rwanda Energy Sector Strategic Plan (2018/19 - 2023/24)

Moreover, on June 2023, the Rwanda Environment Management Authority (REMA) provided to Eni the “Letter of no-objection” to proceed with project activities related to the distribution of Improved Cookstoves. These letters encourage Eni to proceed with the project registration process and the implementation.

2.3.10 Project Ownership

Eni is the owner of the project.

During the distribution of the improved cookstoves to the households, a member for each household signs an End User Agreement transferring to Eni the property rights of the carbon credits generated with the device. Project implementors also inform the users about the necessity of ongoing monitoring process to track the outputs over and above phase.

2.3.11 Grouped Projects

For the inclusion of new project activity instances, the project proponent shall ensure that it meets the eligibility criteria below Grouped Projects (GP) of the SD VISta Standard.

	Criterion	How the new project activity instances to comply
1	<i>Adopt and implement the project activities in the same manner as specified in the project description.</i>	<i>New project activity instances will be implemented in the same manner as described in the project description. At this time, no other project device types will be used.</i>
2	<i>Where appropriate, meet the applicability conditions of the SD VISta asset methodology applied to a project.</i>	<i>The project activity as well as the new project instances do not apply for SD VISta assets and hence this criterion is not applicable.</i>
3	<i>Are subject to the same scenarios at project start with respect to stakeholders' well-being as determined for initial project instance(s), where (per Section 2.1.5 above) the project must meet the criteria of Section 3.1 above</i>	<i>New project activity instances will have the same baseline scenario available at the beginning of the project. Also, the instances will be designed to maximize intended impact and preserve well-being for stakeholders as determined for initial project instances.</i>
4	<i>Are subject to the same scenarios at project start with respect to natural capital and ecosystem services as determined for initial project instance(s) where (per Section 2.1.5 above) projects must meet the criteria of Section 3.2 above.</i>	<i>New project instances will have the same scenario with respect to natural capital and ecosystem services as determined in the baseline scenario.</i>
5	<i>Are subject to the same processes for stakeholder engagement described in the project description</i>	<i>The new project activity instances would occur with the same processes for stakeholder engagement, including, stakeholder consultations, refresher trainings for Implementing Partners and feedback.</i>

6	Are subject to the same processes for respect for rights to lands, territories and resources – including free, prior and informed consent – described in section 2.4 above.	The outlined Project Management (Section 2.3) will apply for each additional project activity.
7	Have similar monitoring elements to those set out in the project description	The new project activity will be subject to equivalent monitoring process, plan and procedures as described in section 3.3.

3 BENEFITS FOR PEOPLE AND PROSPERITY

3.1 Condition of Stakeholders at Project Start

A period of reconciliation and justice began since 2000: Rwanda's economy [24] and Human Development Index have grown rapidly. (Figure 10); between 2006 and 2011 the poverty rate reduced from 57% to 45%,[25] while life expectancy rose from 46.6 years in 2000[26] to 65.4 years in 2021.[27]

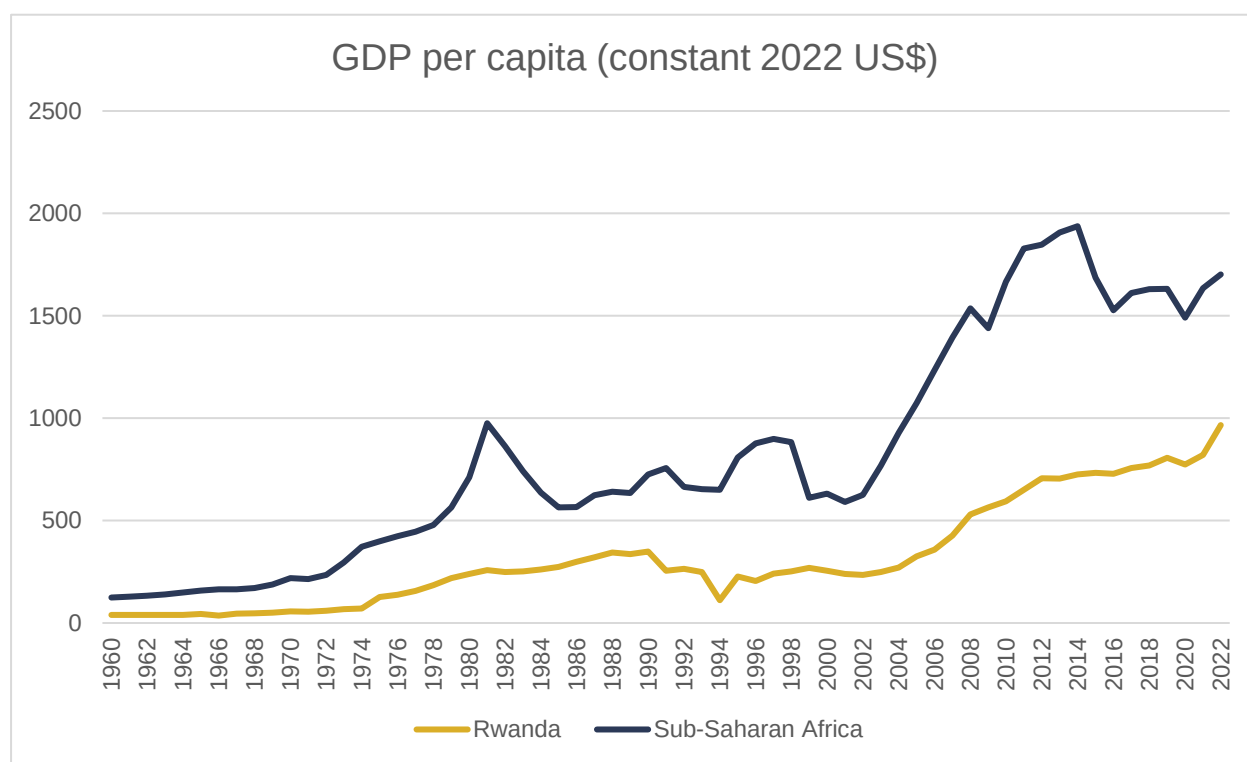


Figure 10: GDP per capita (constant 2022 US\$) - Sub-Saharan Africa, Rwanda (source: World Bank)

a. Households involved in project and other potential people:

Nowadays Rwanda is divided into five provinces (*intara*), named according to geographical exposure (Norther, Eastern, southern, Western and Kigali). The provinces of Rwanda are further divided in 30 district, one of which consists of the capital city Nyagatare. The provinces are further divided into 416 sectors, encompassing smaller localities and aggregated villages.

The latest report of the Global Multidimensional Poverty Index [28] showed that 55.5% of the population lives below the poverty threshold established globally by UN agencies (\$1.90 at purchasing power parity).

Despite recent attempts to improve infrastructure, the economy is still based on the export of products from tea and coffee plantations, where most of the country's workforce is employed, while industry is still underdeveloped.

Regarding the female condition, according to Rwanda Demographic and Health Survey women represent the largest number of poor people in Rwanda.

The former languages spoken in Rwanda are Kinyarwanda, English and French. A small part of population also speaks Swahili, a Bantu language widespread in many areas of eastern and central-southern Africa.

Rwanda's literacy rate for 2021 was 75.9%, and Education expenditures are 4.75% of GDP.

According to the World Bank database, in Rwanda in 2021, only 5% of population have access to clean fuel and technologies for cooking. Firewood and charcoal are the main energy sources especially in rural areas; women and children, especially girls, are usually responsible for the collection and cooking activities of the household. With the traditional and low-efficient cooking system, women and children are exposed to polluting air for an extended duration, causing high levels of respiratory illness. Moreover, due to the long cooking times on the inefficient system, much of the nutrients in the food are lost resulting in the consumption of lower nutrient-rich food. According to IEA Africa Outlook 2022 [30], to bridge the gap and achieve full access to clean cooking for all, liquefied petroleum gas (LPG) is the most scalable solution for urban settlements, while improved biomass cookstoves do most to provide access in rural areas.

Rwanda faces challenges in its switch to modern clean cooking methods, one reason being the low personal income that cannot cover the electricity tariff for electric cookstoves.

Indeed, as highlighted by baseline and KPT report conducted over the people affected by the project, 49% of people use three – stone fire system for cooking, 43 % of people use traditional cookstove (32% wood, 11% charcoal) and only 8% people use and improved cookstove (2% wood, 4% charcoal and 2% LPG) for cooking.

The baseline and KPT report also under light that 70% of respondent intend to use an ICS for several reason; by these, 54% of respondent reported that this choice is driven by fuel saving and cost reduction.

Considering what mentioned before, the market for efficient clean cookstoves is growing in Rwanda, representing the chance to own a cost-effective solution of switching the rural low-income and urban households to modernized cooking methods, reducing their fuel uses and costs [20]

Considering that, as highlighted by the Baseline report, 86% of the person in charge of cooking is represented by woman and children, which are exposed to high risk of respiratory disease and health problem (physical and medical) due to traditional cooking devices, and face numerous risk during fuel collection (animals, dangerous terrain, conflicts, ...) the development of a project to ensure clean cooking access in these areas can lead to significant changes in lifestyle of local population.

With the baseline report it was possible to state that the entire rural population taking part in the project is aware of the risks and harmful effects caused by adopting traditional cooking systems (deforestation, pollution, health issues for lungs, throat, eyes and skin); by this, 87% are aware of most of the negative effects, 14% are aware of a limited part. Moreover, more than half of the surveyed population is also aware that their cooking system can cause risks and dangers.

They are also aware about the positive effect of clean improved cookstoves; indeed, as highlighted by the report, nearly 70% of the respondent is aware about the benefits to switch from traditional cooking system to an efficient improved cookstove.

These considerations lead to the assumption that there may be a strong penetration of technology and that it may be used, understood and positively valued by the local community, improving their livelihood.

All these parameters will be monitored as the project progresses.

b. Implementing Partner

RWN has been chosen as optimal field implementing actor for several reasons: first, his deep knowledge of the local community habits guarantee a high degree of acceptability of the device by families; in addition, due to their knowledge of the territorial roots and the capillary network available to the association make it possible to guarantee effective and efficient distribution. Moreover, once the stove will be locally produced, RWN will be able to support local producers by fine-tuning the local cookstove model via technical support. Finally, the very purpose of the association to contribute to mitigation actions by supporting local employment, in particular female employment, fully adheres to Eni's strategies of attention to local content.

3.2 Expected Impacts on Stakeholders

Impact #1	<i>Access to clean cooking technology</i>
Type of Impact	<i>Positive, actual, direct</i>

Affected Stakeholder Group(s)	<i>Improved cookstoves users (people) and their families</i>
Resulting Change in Well-being	• <i>Reduce burden risk</i> • <i>Reduce time to prepare food</i> • <i>Reduce time to collect fuel</i> • <i>Save income</i> • <i>Reduce exposure to pollutant emissions</i> • <i>Increase food security due to nutrient retention with decreased cooking time</i>
Impact #2	<i>Better health condition</i>
Type of Impact	<i>Positive, actual, direct</i>
Affected Stakeholder Group(s)	<i>Improved cookstoves users (people) and their families</i>
Resulting Change in Well-being	• <i>Reducing users' exposure to pollutant emissions due to higher efficiency of combustion</i> • <i>Reducing burn risk for women and children due to enclosure fire in the combustion chamber</i>
Impact #3	<i>Time saving</i>
Type of Impact	<i>Positive, predicted, direct</i>
Affected Stakeholder Group(s)	<i>People, mostly women and children, in charge of cooking activities and collecting fuel activity</i>
Resulting Change in Well-being	• <i>Reducing women cooking activities time, possibility to invest their predicted saved time in income generating activities</i> • <i>Reducing children domestic work, freeing up predicted saved time for education</i>
Impact #4	<i>Income saving</i>

Type of Impact	<i>Positive, predicted, direct</i>
Affected Stakeholder Group(s)	<i>Improved cookstoves users (people) and their families</i>
Resulting Change in Well-being	• <i>Reducing money for wood or charcoal purchase</i> • <i>Increasing household income and purchasing power for basic necessities (food, medications,...); increasing money for invest in education or in income generating activities</i>

Impact #5	<i>Increase cookstove locally produced and distributed</i>
Type of Impact	<i>positive, predicted, direct</i>
Affected Stakeholder Group(s)	<i>Cookstove manufacturer and employees</i>
Resulting Change in Well-being	• <i>Increase job opportunities in cookstove supply chain</i> • <i>Increase job opportunities in project monitoring and maintenance activity</i> • <i>Expansion of the production capacity</i>

Impact #6	<i>Decrease fuelwood and charcoal sale</i>
Type of Impact	<i>Negative, predicted, indirect</i>
Affected Stakeholder Group(s)	<i>Local vendors</i>
Resulting Change in Well-being	<i>Reducing in demand of wood fuel associated with the efficiency of the improved cookstoves device</i>

Impact #7	<i>Training on project implementation activities</i>
Type of Impact	<i>Positive, Predicted, Indirect</i>

Affected Stakeholder Group(s)	<i>Implementing Partner, Monitoring staff</i>
Resulting Change in Well-being	• <i>Sharing skills related to community engagement, survey conduction</i> • <i>Sharing technical skills related on how conducting Water Boiling Test</i>

3.3 Stakeholder Monitoring Plan

To increase the accountability rate of the project with respect to the achievement of the SDGs, the contributions have been split into “Direct” and “Indirect” effects of the project activities.

The “Direct” contributions are monitored through quantitative analysis and surveys; they are related to the SDGs number 1,3,5,7,8,13. The “indirect” contributions are supported by qualitative and literature research in the area and are not directly measured on the field: they are the SDGs number 2, 3, 10, 15, 17.

The Monitoring Plan represents a key element to improve the current and future management of the outcomes of the stakeholder engagement process. The Project Proponent systematically monitors and evaluate the level of the project’s objectives for stakeholders.

The data and information collected about the project are used to guarantee:

- The project progress state check
- The planning of any corrective measures to be applied during the implementation phase
- The elements for evaluating the effectiveness of implemented measures and the long-term impacts derived from them for the stakeholders.

The analysis and classification of the collected information from stakeholders may lead to a request for corrective measures or additional data.

The stakeholder impacts will be evaluated through the assessment of the primary survey associated with the implementation of project activities.

To increase the accountability rate of the project with respect to the achievement of the SDGs, the contributions have been split into “Direct” and “Indirect” effects of the project activities. The following table describes the KPI of the project to account for the contribution to the achievement of the SDGs: the SDG indicators related to the “Direct” effects of the projects are monitored through quantitative measurements, while the SDG indicators related to the “Indirect” effects are evaluated through qualitative assessment or literature research as described in the sections above.

<i>Project Elements</i>	<i>SDG Indicators</i>	<i>STK Group</i>	<i>Project KPI</i>	<i>Parameter description</i>	<i>Source</i>	<i>Sampling</i>
<i>Project promotes the adoption of improved technology</i>	<i>7.1.2 Proportion of population with primary reliance on clean fuels and technology</i>	<i>Households</i>	<i>Number of improved cookstoves of both types used and in good condition operating in year y</i>	<i>The parameter describes the progression in cookstove distribution, proving that each year there are cookstoves distributed and operating in the same area where, during baseline, such devices were not used.</i>	<i>Improved cookstoves distribution records</i>	<i>100 per cent of data: the progress of the cookstove distribution activity is followed step by step through the use of a mobile App tailored for the project, which allows each distributor to register the contract, the code of the stove delivered and associate it with the residence family with specific GPS coordinates.</i>
<i>Project improves the access to basic services</i>	<i>1.4.1 Proportion of population living in households with access to basic services.</i>					
<i>Project reduces households expenses for purchase charcoal and woodfuel</i>	<i>1.1.1 Proportion of the population living below the international poverty line by sex, age, employment status and geographic location (urban/rural)</i>	<i>Households</i>	<i>Saved budget for fuel for cooking activities</i>	<i>Qualitative assessment to find out the people' perceptions about the improvement of households' budgets and savings due to cookstove utilization</i>	<i>Households baseline and monitoring surveys to assess the people perception about their money and time saving.</i>	<i>Simple random sampling among people in the target area</i>
<i>Project targets vulnerable people</i>	<i>10.2.1 Proportion of people living below 50 per cent of median income, by sex, age and persons with disabilities</i>					

<i>Project Elements</i>	<i>SDG Indicators</i>	<i>STK Group</i>	<i>Project KPI</i>	<i>Parameter description</i>	<i>Source</i>	<i>Sampling</i>
<i>Project enables the development of a local supply chain</i>	<i>8.2.1 Annual growth rate of real GDP per employed person.</i>	<i>Cookstoves manufacturer/ employees</i>	<i>Number of ICS locally produced</i>	<i>Amount of cookstove locally produced or assembled per year</i>	<i>ICS purchasing records</i>	<i>100 per cent of data: the progress of the cookstove production is followed step by step in order to organize the distribution activity</i>
<i>Project contribute to increase local employment</i>	<i>8.5.2 Unemployment rate, by sex, age and persons with disabilities</i>	<i>Local employees</i>	<i>Number of local people directly involved in project activities.</i>	<i>Amount of local people involved in project activities, such as distribution and monitoring activities, compared to the start of the project</i>	<i>HR Records</i>	<i>100 per cent of data: the progress of employment is updated to follow the plan of distribution and production</i>

<i>Project Elements</i>	<i>SDG Indicators</i>	<i>STK Group</i>	<i>Project KPI</i>	<i>Parameter description</i>	<i>Source</i>	<i>Sampling</i>
<i>Project reduces women unpaid domestic work</i>	<i>5.4.1 Proportion of time spent on unpaid domestic and care work, by sex, age and location</i>	<i>Women</i>	<i>Reduced amount of fuel needed for cooking activities</i>	<i>Measurement on field of biomass used by each people' family for cooking activity respect to the situation previous the starting of the project</i>	<i>Monitoring Report for the corresponding VCS Project</i>	<i>Simple random sampling with 90 per cent confidence interval and a 10 percent margin of error achieved</i>
			<i>Increased time available to do other activities other than cooking and collecting/buying fuel.</i>	<i>Qualitative assessment to find out the women perceptions about their available time due to cookstove utilization compared to the situation previous to the starting of the project</i>	<i>Households baseline and monitoring surveys</i>	<i>Simple random sampling among people in the target area</i>
<i>Project reduces time needed for cooking activities</i>	<i>2.1.1 Prevalence of undernourishment</i>	<i>Households</i>	<i>Saved time for cooking activities</i>	<i>Relation between adoption of cleaner devices for cooking activities and increased dietary diversity and consumption</i>	<i>Surveys</i>	

<i>Project Elements</i>	<i>SDG Indicators</i>	<i>STK Group</i>	<i>Project KPI</i>	<i>Parameter description</i>	<i>Source</i>	<i>Sampling</i>
<i>Tackle the issue of air pollution by reducing emissions and creating awareness</i>	<i>3.9.1: Mortality rate attributed to household and ambient air pollution</i>	<i>Women, households</i>	<i>Reduced amount of fuel needed for cooking activities</i>	<i>Measurement on field of biomass used by each people' family for cooking activity respect to the situation previous the starting of the project</i>	<i>Monitoring Report for the corresponding VCS Project</i>	<i>Simple random sampling with 90 per cent confidence interval and a 10 percent margin of error achieved</i>
			<i>Reduced amount of injuries (respiratory and eyes disease) caused by smoke in cooking activities</i>	<i>Qualitative assessment to find out the people' perceptions about the improvement of their health conditions (respiratory and eyes disease) linked to the exposure to the combustion smoke during cooking activities</i>	<i>Households baseline and monitoring surveys</i>	<i>Simple random sampling among people inf the target area</i>
			<i>Reduced risk of safety exposure in cooking activities</i>	<i>Qualitative assessment to find out the people perceptions about the improvement of their safety</i>	<i>Households baseline and monitoring surveys</i>	<i>Simple random sampling among people inf the target area</i>

<i>Projects reduces risks of domestic accidents</i>	<i>3.2.1. Under 5 morality rate</i>			<i>conditions and risks exposure linked to cooking activities</i>		
---	-------------------------------------	--	--	---	--	--

3.4 Net Positive Stakeholder Well-being Impacts

The expected impacts defined in section 3.2 are positive for each stakeholder group with the exception of the impact on “wood/charcoal vendors” that is predicted to be negative. However, it is likely that the reduction in the demand for wood/charcoal within the project boundary will push the vendors to sell the fuel outside the project area. Indeed, 99.3% [9] of the rural population in Rwanda still rely on the traditional system for cooking and the project impact will not cause a significant reduction in the business. Moreover, the local vendors represent a fraction of the population: therefore, the predicted negative impact may be compensated by the long-term sustainable development benefits observed:

- A higher section of the population will have access to basic services and will rely on clean cooking technology;
- People and their families will improve their health status: fewer individuals will be affected by ambient air pollution and fewer children will be a risk of experiencing burns and injuries associated with open cooking fires;
- People and their families will improve their status to approach a food security level due to nutrient retention with decreased cooking time;
- People, especially women and children, will spend a lower portion of time on unpaid domestic and care work and can spend time in more productive and income-generating activities or simply resting or studying;
- People will have a greater capacity to save money due to reduced demand for fuel;
- A higher proportion of the population will have access to training for financial education, savings programs, and long-term employability skills.

4 BENEFITS FOR THE PLANET

4.1 Condition of Natural Capital and Ecosystem Services at Project Start

Rwanda borders Burundi, Democratic Republic of Congo, Tanzania and Uganda. Rwanda has a temperate tropical highland climate, with lower temperatures than are typical for equatorial countries because of its high elevation.

The country has four primary climatic regions: eastern plains, central plateau, highlands, and regions around Lake Kivu. The eastern plains receive an annual rainfall of between 700 mm and 1,100 mm, with mean annual temperature oscillating between 20°C and 22°C.

The central plateau region enjoys rainfall of between 1,100 mm and 1,300 mm, with an annual mean temperature of between 18°C and 20°C.

The highlands, including the Congo-Nile Ridge and volcanic chains of Birunga, benefit from an annual rainfall of between 1,300 mm and 1,600 mm and experience annual mean temperatures between 10°C and 18°C.

Regions around Lake Kivu and Bugarama plains get annual rainfall of between 1,200 mm and 1,500 mm with annual mean temperatures between 18°C and 22°C. [39]

The rainfall patterns are characterized by four seasons, a short rainy season from September to November and a longer season between March and May. Between these seasons are two dry periods, a short one between December and February and a long one from June to August. [40]

Rwanda is vulnerable to natural disasters due to its geographical location (East African Rift Valley, near Nyiragongo volcanoes) and the effect of the Inter Tropical Convergence Zone (ITCZ) on the continent. Such disasters are exacerbated by poor farming practices, deforestation and environmental degradation among others. [40]

Rwanda has a diversified natural ecosystem ranging from afro-montane in the Northern and Western Provinces to lowland forests, savannah woodlands, savannah grasslands in the Southern and Eastern Provinces. Rwanda is also rich in large numbers of inland freshwater lake and wetland ecosystems. These ecosystems provide a wide range of goods and services, but they are under pressure from human activities. The most critical ecosystems include forests, water, wetland ecosystems, among others. Some of these have been already converted into agricultural use and settlements. [41]

Rwandan forests area were estimated at 240,746 ha in 2007. This translates to approximately 10 per cent of the surface of the national dry lands (23,835 km²).

However, forest resources in Rwanda include also small woodlots and agro-forestry resources.

In Rwanda, humid natural forests constitute the majority of the forest cover (33%), followed by Eucalyptus plantations and degraded natural forests at 26% and 15.7% respectively.

Statistics from the Forest department show that forests were estimated to cover 659,000 ha in 1960 (ROR 2004). This reflects a loss of approximately 64% of forests in between 1960 and 2007, which is more than 1.3% per cent per year. The rapid increase in population is increasing pressure on forests in terms of encroachment and deforestation.

Rwanda has four types of protected areas which includes national parks (Akagera, Nyungwe and Volcanoes National Park); forest reserves (Gishwati, Iwawa Island and Mukura Forest reserves); forests of cultural importance (Buhanga forest); and wetlands of global importance (Rugezi-Bulera-Ruhondo wetland complex). Besides those forests with a legal status of protected areas, there are other forests of cultural importance (Busaga forest in Muhanga district) and other remnant natural forests which are protected by law. In fact, the current law on forests prohibits human activities in natural forests. [42]

4.2 Expected Impacts on Natural Capital and Ecosystem Services

Impact #8	<i>Reduction GHG emission</i>
Type of Impact	<i>Positive, predicted, indirect</i>
Affected Natural Capital and/or Ecosystem Service(s)	<i>Air quality in the households of the people</i>
Resulting Change in Condition	<i>The improved cookstoves require less fuel to produce the same amount of thermal energy compared to a traditional cooking system: therefore, there is a reduction of emission related to the cooking activity, leading to a positive impact on the environment</i>

Impact #9	<i>Reduction fuel wood and charcoal consumption</i>
Type of Impact	<i>Positive, actual, direct</i>
Affected Natural Capital and/or Ecosystem Service(s)	<i>Forest ecosystem: live plants, biodiversity and species living in the system</i>
Resulting Change in Condition	<i>By using the improved cookstoves, the amount of fuel for cooking purposes decrease given the better energy efficiency of the device: this leads to a drop in the wood and charcoal demand and, therefore,</i>

a decrease in the degradation and deforestation rates since the logging activity is one of the main drivers.

4.3 Natural Capital and Ecosystem Services Monitoring Plan

According to the VCS project for the corresponding SD VISTA period, the net GHG emission reductions are computed applying Equations of the “VRM0006” v1.2 methodology, for both firewood and charcoal improved cookstoves replacing baseline cooking systems (three-stone fire or rudimentary stoves). The methodology “VMR0006” v1.2 does not compute baseline emissions but quantifies emission reductions “ $ERY_{i,j}$ ” directly as a function of the reduction of non-renewable woody biomass consumption ($By,savings_{i,j}$) resulting from the sustained use of the improved cookstoves distributed by the project in substitution of the traditional cooking methods.

<i>Project Elements</i>	<i>SDG Indicators</i>	<i>Natural ecosystem impacted</i>	<i>Project KPI</i>	<i>Parameter description</i>	<i>Source</i>	<i>Sampling</i>
<i>Project contributes to lower GHG emissions</i>	<i>13.2.2 Total greenhouse gas emissions per year</i>	<i>Atmosphere</i>	<i>Reduced amount of fuel needed for cooking activities</i>	<i>Measurement on field of biomass used by each</i>	<i>Monitoring Report for the corresponding VCS Project</i>	<i>Simple random sampling with 90 per cent confidence interval and a 10 percent margin of error achieved</i>
<i>Project contributes to reduce pression on deforestation</i>	<i>15.1.1 Forest area as a proportion of total land area</i>	<i>Forest ecosystem</i>	<i>Reduced deforestation rate in project area</i>	<i>Relation between the use of more efficient devices for cooking and the amount of wood collection in forest not sustainably managed</i>	<i>Evidence in the reduction of the woody biomass for cooking purposes</i>	
	<i>15.2.1 Progress towards sustainable forest management</i>		<i>Increased sustainable management of the forest in project area</i>	<i>Relation between the use of more efficient devices and the progress in forest conservation and sustainably managed</i>	<i>Evidence in the reduction of the woody biomass for cooking purposes</i>	
<i>Project may increase awareness on climate change</i>	<i>13.3.1 Improve education on climate change mitigation and adaptation.</i>	<i>At global level</i>	<i>Attendance to behavioral change campaign</i>	<i>Direct engagement and attendance of people to behavioral change campaigns</i>	<i>Attendance of people to behavioral change</i>	

For SDG13, the parameters are estimated through the equations consistent with the methodology for the corresponding VCS projects ID 3092 as follow

Monitoring Parameter	Equation	Source	Sampling
Reduced GHG emission	$ER_y = \sum \sum B_{y,savings,i,j} \times N_{0,i,j} \times n_{y,i,j} \times \mu_y \times f_{NRB,y} \times NCV_{biomass} \times (EF_{wf,CO2} + EF_{wf,non\ CO2}) \times Adj_{LE} \times (1 - u_d)$	Monitoring Report for the corresponding SD VISta period of VCS Project	Simple random sampling with 90 per cent confidence interval and a 10 percent margin of error achieved
Fuelwood saved by the project improved cookstove utilization	$B_{y,savings,i,j} = B_{old,wood} - B_{new,KPT,i,j}$	Monitoring Report for the corresponding SD VISta period of VCS Project	Simple random sampling with 90 per cent confidence interval and a 10 percent margin of error achieved

Where:

i = Indices considering more than one type/model of improved cookstove introduced to replace baseline cooking technique (three-stone fire or traditional brazier).

j = Indices considering more than one batch of improved cookstove of type i issued in the project implementation.

ER_y = Emission reductions by improved cookstoves during year y (tCO_{2e}).

B_{y,savings,i,j} = Quantity of woody biomass saved per improved cookstove of type i and batch j (tonnes).

N_{0,i,j} = Number of project devices of type i and batch j commissioned (number)

n_{y,i,j} = Proportion of commissioned project devices of type i and batch j (N_{0,i,j}) that remain operating in year y (fraction)

μ_y = Adjustment to account for any continued use of pre-project devices during the year y

f_{NRBy} = Fraction of woody biomass that can be established as non-renewable biomass (%).

NCV_{biomass} = Net calorific value of the non-renewable woody biomass that is substituted or reduced (IPCC default: 0.0156TJ/tonne).

$EF_{\text{wf},\text{CO}_2}$ = CO_2 emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel: 112 tCO₂/TJ).

$EF_{\text{wf},\text{nonCO}_2}$ = Non-CO₂ emission factor for the use of wood fuel in baseline scenario (IPCC default for wood fuel: 9.46 tCO₂/TJ).

Adj_{LE} = Adjustment factor to account for leakage related to the non-renewable woody biomass saved by the project activity (fraction)

u_d = Uncertainty deduction for f_{NRB} (%)

$B_{\text{new},\text{KPT},i,j}$ = Annual quantity of woody biomass used in tonnes per project device of type i and batch j, measured as per the KPT protocol (tonnes/year)

It is important to underline that each claim on carbon credits will be carried out through the ex-post calculation, performing the KPTs on a proper sample of ICSs beneficiary households in order to evaluate the effective woody biomass savings on field.

For the GP described, only one improved cookstove will be issued for each household, according to their current cooking fuel (firewood or charcoal). Therefore, biomass saved will be associated to the use of just one device per household, with no necessity to take into account any woody biomass apportions between multiple improved cookstoves.

The monitoring plan design is based on the CDM Guideline “Sampling and surveys for CDM project activities and programmes of activities” v4.0

Sampling Design

Sampling activities are defined at the Grouped Project-level, hence these guidelines will be applied for each Project Instance included in the GP, either individually or in groups

In accordance with the Standard, the parameters whose confidence precision is not explicitly mentioned in the methodology VMR0006 v 1.2 shall be evaluate with 95/10 confidence precision level since the GP is categorized as “large scale” on CDM base.

In case the required confidence precision level is not met, the lower bound of the confidence interval of the parameter value may be chosen as an alternative to repeating the survey efforts.

Parameter values may be estimated by sampling in accordance with Sampling Standard for a single or a group of PI. If required, a single sampling plan covering more PIs will be justified when either the homogeneity of parameters of interest can be demonstrated or the difference among the PIs is considered in the sample size calculation.

The target population will include ICS recipient households by PI in the monitoring database.

For each survey, the sampling method is determined using a random approach on the total ICS people chosen from the monitoring database and based on the precision requirements listed above.

The sample size for each monitoring activity will be the total number of ICS distributed and included in the monitoring database. The sample size will be calculated through a calculation.

Oversampling will be conducted to ensure that the required level of precision is met. The formula used for sampling is the following:

- For proportion-based parameters (i.e. number of operational improved cookstoves):

Where:

$$n \geq \frac{1,645^2 \times N \times p(1-p)}{(N-1) \times 0,1^2 \times p^2 \times 1,645^2 \times p(1-p)}$$

n Sample size

N Total number of households involved in clean cooking project

p Expected proportion

1,645 Coefficient that represent the confidence level at 90%

0,1 Coefficient that represent the precision level at 10%

- For mean-based parameters (i.e. woody biomass used with improved cookstoves):

$$n \geq \frac{1,645^2 \times N \times V}{(N-1) \times 0,1^2 \times p^2 \times 1,645^2 \times V}$$

Where:

V Standard deviation: $\left(\frac{SD}{mean}\right)^2$

n Sample size

N Total number of households involved in clean cooking project

Mean Expected mean value

SD Expected standard deviation value

1,645 Coefficient that represent the confidence level at 90%

The Sampling Frame will be the monitoring database which may contain information from all PI.

Data Recording

Field-based monitoring activities to determine monitored parameter values could be a field survey, KPT.

Data shall be recorded for each activity either directly onto a smartphone-based app designed for the purpose, or, where this is not possible, via paper forms.

Once collected, data will be stored in a monitoring database.

Quality Assurance/Quality Control

In order to minimize sampling errors, over-sampling will be considered, enlarging the sample size in order to ensure a number that meets the required level of precision. This will ensure the integrity of the sample group and maintains the randomness of participant selection. All sample groups will be re-selected for each monitoring period.

If the required level of precision is not achieved, then further surveys may be implemented, or the lower bound value could be adopted in the calculations.

Data analysis

Completed survey questionnaires and KPT data will be collected and analysed by Project Proponent. After a rigorous cross-check for potential errors and mistakes during the field activities, the monitoring report will be prepared. The monitoring report will include:

- Summary of activities undertaken and main information to explain the context (i.e. location, date, description of activities implemented)
- Summary of data collected and values
- Calculation of the parameter values to be utilized in emissions reduction calculations

Implementation Plan

The implementation of the monitoring plan will be the full responsibility of Project Proponent and their partner as implementing actors (i.e. RWN). The monitoring team will be trained in a proper way to develop all the necessary activities and it will be comprised of local people who speak the local languages and are fully aware of local traditions and habits. Individuals will be selected based on their competence and experience for each monitoring activity and trained accordingly.

To prevent any conflict of interest, there will be no incentives provided for these individuals for the type of data provided.

4.4 Net Positive Natural Capital and Ecosystem Services Impacts

Replacing traditional cooking methods with improved cookstoves technology is going to generate a decrease in polluting emissions. The GHG emissions of this project have been estimated assuming an initial number of improved cookstoves planned for the first Project Instance equal to 40,000. The devices' parameters declared by local stoves manufacturer, such as the thermal efficiency and the expected lifespan, have been certified by the Rwanda Standard Board, a national body whose purpose is to establish and publish national standard.

For the ex-ante calculation, the devices that have already reached the expected lifetime are considered as non-operating, even if some of them could still be working. In addition, considering the possible loss of operational devices even before the end of their lifespan a decreasing operation rate ("drop off rate") of 5% per year has been assumed and included in the emission reduction calculation for the 10-year crediting period.

Within this framework, the average annual CO₂ emission reduction expected, resulting from the implementation of this clean cooking project, is around 37,628 tCO₂e /year (almost 0.37 MtCO₂e in the entire project's lifetime).

The total net positive ecological impact includes not only the emission reductions related to the cooking activity, but also the decreasing deforestation rate due to the energy efficiency of the new device.

No negative ecological impacts have been identified in this project; therefore, net impacts of the activities are positive.

APPENDIX

Use appendices for supporting information. Delete this appendix (title and instructions) where no appendix is required.

References:

1. Kammila, S., Kappen, J. F., Rysankova, D., Hyseni, B., & Putti, V. R. (2014). Clean and improved cooking in Sub-Saharan Africa : a landscape report. ACCES, 1–182.
<http://documents.banquemondiale.org/curated/fr/164241468178757464/Clean-and-improved-cooking-in-Sub-Saharan-Africa-a-landscape-report>
2. IMPROVING THE EFFICIENCY AND SUSTAINABILITY OF CHARCOAL AND WOODFUEL VALUE CHAINS
May 31, 2022
<https://documents1.worldbank.org/curated/en/099845106292213741/pdf/P163358084c95e020b5280fba571d072a2.pdf>
3. Mortality rate attributed to household and ambient air pollution – Rwanda, 2019.
<https://data.worldbank.org/indicator/SH.STA.AIRP.P5?locations=RW>
4. THE STATE OF GENDER EQUALITY IN RWANDA FROM TRANSITION TO TRANSFORMATION GENDER MONITORING OFFICE MARCH | 2019. https://africa.unwomen.org/sites/default/files/2023-11/rwanda_gender_equality_profile_0.pdf
5. ESMAP. The State of Access to Modern Energy Cooking Services (English). Washington, D.C. : World Bank Group. <http://documents.worldbank.org/curated/en/937141600195758792/The-State-of-Access-to-Modern-Energy-Cooking-Services>
6. CASA Consulting. (2001). Evaluation of the Dadaab firewood project, Kenya. UNHCR.
7. IEA, A vision for clean cooking access for all, (Paris, 2023) <https://www.iea.org/reports/a-vision-for-clean-cooking-access-for-all>
8. Access to clean fuels and technologies for cooking, Rwanda, World Bank.
<https://data.worldbank.org/indicator/EG.CFT.ACCS.ZS?locations=RW>
9. Access to clean fuels and technologies for cooking, Rwanda, World Bank.
<https://data.worldbank.org/indicator/EG.CFT.ACCS.ZS?locations=RW>
10. Nzungu, Daniel & Maina - Phd, Paul & Njeru, Chrocosiscus. (2021). Barriers to the Adoption of Improved Cooking Stoves for Rural Resilience and Climate Change Adaptation and Mitigation in Kenya. 10.1007/978-3-030-42091-8_133-1.
11. World Food Programme, Clean and modern energy for cooking. A path to food security and sustainable development, (Rome, May 2022)
12. EnDev 2016: Empowering People – Report on Impact. GIZ- Eschborn.
p.15 <https://endev.info/content/Downloads>
13. Bwenge, Neatness Stanley 2011: The Effects of Adopting Improved Wood Stoves on the Welfare of Rural Women: A case of Kibaha District in Tanzania. Van Hall Larenstein University of Applied Sciences, Netherlands. P.27. <http://edepot.wur.nl/192647>
14. UNDP 2013: Gender and energy – p.26 <http://bit.ly/2vuZHKy>
15. Word Bank
<https://www.worldbank.org/en/country/rwanda/overview#:~:text=Rwanda's%20economy%20has%20remained%20resilient,projected%20average%20growth%20of%207.2%25>
16. UNDP, Social and Environmental Standards: Stakeholder engagement guidelines (New York, 2020)
17. Chrétien, Jean-Pierre (2003). The Great Lakes of Africa: Two Thousand Years of History. Cambridge, Massachusetts: MIT Press. ISBN 978-1-890951-34-4. Archived from the original on 11 January 2023. Retrieved 25 June 2019.

18. Gourevitch, Philip (2000). *We Wish To Inform You That Tomorrow We Will Be Killed With Our Families* (Reprint ed.). London; New York, N.Y.: Picador. ISBN 978-0-330-37120-9. Archived from the original on 11 January 2023. Retrieved 25 June 2019.
19. Prunier, Gérard (1995). *The Rwanda Crisis, 1959–1994: History of a Genocide* (2nd ed.). London: C. Hurst & Co. Publishers. ISBN 978-1-85065-243-4.
20. Nyaga, H. Njiru, J. & Baranda Alonso, I. Ndayishimiye, and D. Ntivunwa Saulve. "Policy and Market Review for Modern Energy Cooking in Rwanda." *Modern Energy Cooking Services*, June 22, 2022. <https://mecs.org.uk/publications/policy-and-market-review-for-modern-energy-cooking-in-rwanda>
21. Henley, Jon (31 October 2007). "Scar tissue". *The Guardian*. London. Archived from the original on 1 September 2013. Retrieved 13 November 2015.
22. Dallaire, Roméo (2005). *Shake Hands with the Devil: The Failure of Humanity in Rwanda*. London: Arrow. ISBN 978-0-09-947893-5. Archived from the original on 1 July 2023. Retrieved 25 June 2019.
23. Bowcott, Owen (2 April 2014). "Rwanda genocide: the fight to bring the perpetrators to justice". *The Guardian*. London. Archived from the original on 20 February 2020. Retrieved 12 April 2017.
24. World Bank. "GDP (current US\$), graph". Archived from the original on 28 June 2016. Retrieved 19 August 2015
25. National Institute of Statistics of Rwanda (February 2012). "The third Integrated Household Living Conditions Survey (EICV 3) – Main indicators Report" (PDF). Archived from the original (PDF) on 20 November 2015. Retrieved 19 November 2015.
26. United Nations Development Programme (UNDP) (V) (2013). "Human Development Report 2013: Rwanda" (PDF). Archived (PDF) from the original on 18 October 2018. Retrieved 31 August 2015.
27. World Bank. "Life expectancy at birth, total (years) – Rwanda"
<https://data.worldbank.org/indicator/SP.DYN.LE00.IN?locations=RW>
28. United Nations (UN), United Nations, Global Multidimensional Poverty Index, 2019 -
<https://hdr.undp.org/content/2019-global-multidimensional-poverty-index-mpi>
29. Rwanda Demographic and Health Survey 2014-15 Final Report National Institute of Statistics of Rwanda Kigali, Rwanda Ministry of Finance and Economic Planning Kigali, Rwanda Ministry of Health Kigali, Rwanda The DHS Program ICF International Rockville, Maryland, USA March 2016
30. IEA Africa Outlook 2022 (Paris, 2022)
31. "Outreach Programme on the Rwanda Genocide and the United Nations". 12 October 2023.
32. Stroh, Alexander (2010). "Electoral rules of the authoritarian game: undemocratic effects of proportional representation in Rwanda". *Journal of Eastern African Studies*.
33. Matfess, Hilary (2015). "Rwanda and Ethiopia: Developmental Authoritarianism and the New Politics of African Strong Men". *African Studies Review*.
34. Waldorf, Lars (2017). "The Apotheosis of a Warlord: Paul Kagame". In Themnér, Anders (ed.). *Warlord Democrats in Africa: Ex-Military Leaders and Electoral Politics*
35. *Politics in Rwanda | About Rwanda*. www.aboutrwanda.com. Archived from the original on 5 March 2016. Retrieved 12 November 2015.
36. https://www.rema.gov.rw/info/details?tx_news_pi1%5B%40widget_0%5D%5BcurrentPage%5D=5&tx_news_pi1%5Baction%5D=detail&tx_news_pi1%5Bcontroller%5D=News&tx_news_pi1%5Bnew

[s%5D=123&cHash=8286928a1a700d1bd711f2e80fcf4759#:~:text=Rwanda%20has%20a%20diversified%20](#)

37. <https://www.globalforestwatch.org/dashboards/country/RWA/?category=forest-change>

38. <https://www.rema.gov.rw/soe/chap9.php>

39. https://climateknowledgeportal.worldbank.org/sites/default/files/2021-09/15970-WB_Rwanda%20Country%20Profile-WEB.pdf

40. <https://www.rema.gov.rw/soe/chap9.php>

41. https://www.rema.gov.rw/info/details?tx_news_pi1%5B%40widget_0%5D%5BcurrentPage%5D=5&tx_news_pi1%5Baction%5D=detail&tx_news_pi1%5Bcontroller%5D=News&tx_news_pi1%5Bnews%5D=123&cHash=8286928a1a700d1bd711f2e80fcf4759#:~:text=Rwanda%20has%20a%20diversified%20

42. <https://www.rema.gov.rw/soe/chap6.php#:~:text=Rwanda%20forests%20and%20woodlands%20can,Akagera%20national%20park%20and%20remnants>